Bounded-Arity Shared Decoders and Ramsey-Free Composition for Matrix Transposition

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Abstract

The synthesis of [3] traced the stalemate on whether a multi-tape Turing machine can transpose a $d \times d$ matrix of $\lceil \log_2 d \rceil$-bit entries in $O(L)$ steps, where $L = d^2 \lceil \log_2 d \rceil$ (question (T)), to a structural wall: trace reductions on the machine side produce bounded-fan-in cut cones ($\Delta = O_{k,\varepsilon}(1)$) while discarding the finite transition table that generates them, whereas shared-decoder models retain a shared M-class rule while discarding bounded read arity. Because the two invariants were analyzed separately, arbitrary-set shared decoders blew up to rate $1 - \frac{\log_{|\Sigma|} d}{d^2} \to 1$ at unrestricted arity, two-stage linear composition relied on Hales–Jewett arithmetic progressions and collapsed at $M_1 \approx \log \log d$, and cyclic column-shift symmetry could not distinguish matrix transposition from Stone's $O(L)$-time perfect shuffle.

This paper reunites bounded read arity with decoder sharing and establishes six main kernel-verified results in Lean 4. First (Theorem 3.4), for every alphabet $|\Sigma| \geq 2$ (with no dependence on $|\Sigma|$), any subset $C \subseteq \Sigma^{d \times d}$ transposed outside $\mathcal{E}$ by M shared deterministic column decoders of cell read arity $\Delta \geq 1$ satisfies

$$\log_{|\Sigma|} |C| \;\leq\; d^2 \;-\; \big(d^2 - |\mathcal{E}| - dM\Delta\big) \frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}},$$

giving the first unconditional strictly-below-1 capacity bound on arbitrary nonlinear sets at every constant arity $\Delta \geq 3$ (where [3, Problem 9.4(b)] remained open at rate 1, giving e.g. $\leq \frac{229}{256} + o(1)$ at $\Delta = 3$), the first Lean-verified bound below 1 at $\Delta \geq 2$ (complementing the written $M = 1, \Delta \leq 2, \mathcal{E} = \emptyset$ order bound of [3, Proposition 6.22(b)]), and a positive rate deficit across all $\Delta \geq 1$ whenever $M\Delta = o(d)$ and $|\mathcal{E}| \leq (1-\delta)d^2$. Second (Theorems 4.4 and 5.6), we eliminate Hales–Jewett Ramsey numbers from two-stage linear shared composition: via affine Hilbert cubes over any field F we prove the explicit exact-factorization cutoff $d \leq \max\big(8M_1M_2^2,\ (2M_1)^{2^{M_2}}\big)$ (ruling out $K_2 \circ K_1 = \tau$ up to $M_1 = o(d^{1/2^{M_2}})$ for fixed M_2, including on 3-AP-free Behrend partitions), and over any field with $(d : F) \neq 0$ we combine shifted sumsets, arbitrary-subset Vandermonde pullbacks, sparse annihilating polynomials, and iterated Kővári–Sós–Turán extraction to prove an explicit rank deficit $\Omega\big(b\, d\, (\log d)^{1/(2M_2-1)}\big)$ for all constant M_1, M_2 ($\Omega(b\, d \log d)$ when $M_2 = 1$). Third (Theorems 6.1 and 6.5), anti-diagonal untwisting identifies row-shift-equivariant operators with decoupled tensor slices—packaging the 9/16 and 5/8 random-mask certificate floors and realizing the single-shift 5/8 floor via generic tensor subspaces—while Operator Schmidt Rank across the row/column tensor cut separates the transpose ($\mathrm{rk}_\otimes(\tau) = d^2$) from Stone's shuffle ($\mathrm{rk}_\otimes(S) = 4$ for $d = 2^m \geq 2$), yielding a $t \geq \log_2 d$ pass lower bound on alternating shuffle/row/column pipelines ([3, Problems 9.1 and 9.5]). Fourth (Theorems 7.2 to 7.4), we disprove the $d(d+1)/2$ pairing bound for three-term swap codes asked in [3, Remark 5.6] (attaining dimension $4m(m+1)$ at $d = 3m$, which exceeds $d(d+1)/2$ for $m \in \{1,2,3,4\}$, and $29 > 28$ at $d = 7$, giving $d \in \{3,4,6,7,9,12\}$ over every field), resolve the literal multi-scale interval formulation of [3, Problem 9.3] with exact upper bound $(d^2 + d\beta)/2$ where $\beta = \min\big(\max(1, 2\sum_j M_{A,j}), \max(1, 2\sum_j M_{B,j})\big)$, and isolate the periodic Rademacher/Walsh boundary dichotomy. Finally (Theorems 8.2 and 8.3), we construct native

causal-cone evaluation trees directly from the transition function of `Transpose.TM` on frozen cuts ([3, Problem 9.4]). Question (T) itself remains open; we conclude by isolating the sharp finite and algebraic thresholds to which the series now reduces (T).

Contributions. Hans Brende chose the problem and directed the research. Gemini 4 Carbon (Google DeepMind) developed the mathematics, Lean formalizations and text under his direction.

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1 Introduction: Paper 17's wall and where the two spines meet

1.1 The question (T) and the wall of Paper 17

Let $d \geq 2$, let $b = \lceil \log_2 d \rceil$ be the canonical entry bit-width, and let $L = d^2 b = \Theta(d^2 \log d)$ be the bit-length of a $d \times d$ matrix of b-bit words laid out row by row on the read-only input tape of a deterministic multi-tape Turing machine with k read-write work tapes and a write-only output tape ([5, Definition 2.1]). Question (T)—and its stronger time-space trade-off form, the Tape Coding Conjecture [5, Conjecture 9.1]—asks whether every such machine that outputs the transposed matrix $X^\top$ for all inputs X requires $\omega(L)$ steps; by Harvey and van der Hoeven [13], a negative answer to the $O(L)$-time question implies $\Omega(n \log n)$ integer multiplication on multi-tape Turing machines.

After sixteen papers of reductions and barriers, the synthesis paper *Order, Symmetry and the Swap* [3, Section 4] paused to diagnose why every prior attack had stalled short of (T):

(i) *Motion without data* ([3, Sections 4.2 and 4.3]): single-scale pass counting, trace crossing bounds, and non-abelian commutator invariants saturate at one dyadic scale or $O(1)$ passes once a machine is permitted to interleave scales or code across head visits; even oblivious $O(L)$-time transposers need macroscopic crossings ([8, Theorem 5.5, Corollary 5.6, and Section 10.3]), which a two-tape cascade can supply.

(ii) *Codes without machines* ([3, Section 4.4]): once a computation is compressed to an abstract constant-depth or bounded-degree storage DAG whose nodes may apply *arbitrary, distinct* functions at every tape cell, network coding defeats the $\frac{1}{2}$-rate pairing barrier. Paired-distance separation already exceeds rate $\frac{1}{2}$ ([6, Theorem 7.1, Theorem 7.3, and Corollary 8.2], reaching $(\sqrt{3}-1)^2 - \varepsilon \approx 0.536$ over $\mathbb{F}_{p^2}$) and reaches binary rate $1 - O(\Delta^{-\alpha}) \to 1$ as the degree cap $\Delta \to \infty$ across arbitrary odd-girth separation thresholds ([10, Definition 5.1, Theorem 5.5, and Theorem 6.1], formalized conditionally on an odd-girth linear seed).

(iii) *The missing fourth object* ([3, Sections 4.1 and 4.6]): a real Turing machine has a *fixed finite transition table* δ. When a head sweeps across regularly laid-out data, successive cells are not decoded by d^2 bespoke linear combinations tailored to each matrix coordinate (i, j); they are decoded by the *same* finite-state rule applied to shifted views of the tape.

To turn that observation into theorems, [3, Sections 5 and 6] isolated two algebraic pillars of shared rules on a vector space $\mathrm{Mat}_{d\times d}(F)$: *Order* ([3, Theorem 5.1 and Corollary 5.5], showing that a two-term swap identity $A_1 X B_1 + A_2 X B_2 = X^\top$ forces $\dim S \leq d(d+1)/2$ via triangularization of a matrix pencil over the algebraic closure) and *Symmetry* ([3, Definition 6.3 and Theorem 6.4], showing that an M-class linear shared rule is governed by cyclic column shifts $\mathrm{rot}_c(X)_{i,j} = X_{i,\,j+c}$ and satisfies the translate-pigeonhole bound). Two papers then pushed the two halves of the program forward independently: [11] established macroscopic original-entry capacity and bounded-fan-in cut pullback for adaptive machines, while [9] developed the capacity and multi-stage composition theory of M-class shared decoders.

1.2 Two spines, two complementary blind spots

Stepping back across [3, 11, 9] reveals why neither spine could close (T) alone: each spine preserved one half of what makes a tape sweep special and discarded the other.

The Machine Spine: bounded arity $\Delta = O(1)$ without decoder sharing. Trace compression via TCP(k) ([4, Definition 2.3, Proposition 2.4, and Theorem 7.5]) shows that cutting a k-head walk DAG with an ε-fraction vertex cut reduces residual path length to $D = D(k, \varepsilon) = O_{k,\varepsilon}(1)$, trapping each clean output cell inside a causal cone of bounded read fan-in $\Delta \le b(k+1)^D$ (or $\Delta \le (k+1)^D$ per bit or word-step; conditional on a depth-D source cut B, [11, Lemma H.2, Theorem H.5, and Corollary H.6] transports interior-lane cones across adaptive runs). Yet the moment one forgets that those cones are evaluated by a single transition table δ, two barriers block an unconditional lower bound: (a) abstract separated storage codes reach rate $1 - O(\Delta^{-\alpha}) \to 1$ as $\Delta \to \infty$ ([6, 10]), and (b) any $O(n \log n)$-gate Boolean barrel-shift circuit embeds into a max-degree-4 causal network achieving complete-exchange shift coding with $\Omega(1)$ coding-to-routing gain when vertices apply distinct local gate functions ([5, Propositions 11.2 and 11.3]).

The Shared-Decoder Spine: M-class sharing without bounded arity $\Delta = O(1)$. Conversely, [3, 9] enforced M-class decoder sharing: column c of $X^\top$ is produced by applying one of M shared functions $f_{p(c)}$ to the column-shifted state $\mathrm{rot}_c(X)$. However, in the core shared-decoder predicates of [9], each f_m was allowed *unrestricted* read arity ($\Delta = d^2$). Dropping $\Delta = O(1)$ erected three artificial walls:

Wall 1. Arbitrary-set blow-up at unrestricted arity. Fixing an arbitrary nonlinear subset $C \subseteq \Sigma^{d \times d}$ arises naturally when one freezes a trace cut (partitioning inputs into nonlinear trace/cut fibres). At unrestricted read arity $\Delta = d^2$, a single shared deterministic decoder ($M = 1$) already achieves rate $1 - \frac{\log_{|\Sigma|} d}{d^2} \to 1$ via cyclic-orbit markers ([9, Theorem A.1 and Corollary A.2]). In [3, Section 6.7, Propositions 6.21–6.23, and Problem 9.4(b)], written (unformalized) acyclic-subgraph and order arguments settled bounded offsets ([3, Proposition 6.21(a)]), window one ($\Delta = 1$, [3, Proposition 6.21(b)]), exact 1-class window two ($M = 1, \Delta \le 2, \mathcal{E} = \emptyset$, rate $\le \frac{1}{2} + \frac{3}{2d}$, [3, Proposition 6.22(b)]), and bounded-value offsets or mutual helpers ([3, Proposition 6.22(c)–(d) and Proposition 6.23]). Yet for every constant window $\Delta \ge 3$ with spread rows and offsets ([3, Problem 9.4(b)]), as well as for sublinearly growing class counts $M = o(d/\Delta)$ or nonzero broken-output sets $\mathcal{E} \neq \emptyset$—especially over the canonical word alphabet $|\Sigma_d| = 2^{\lceil \log_2 d \rceil} \ge d$, where one symbol can encode its own shift index c—no nontrivial upper bound < 1 on arbitrary sets was known.

Wall 2. Hales–Jewett collapse in two-stage linear composition. A two-pass machine induces a composition $E_2 \circ E_1 = \tau$ of two shared stages. In [9, Section 6, Lemma 6.5, and Theorem 6.6], bounding the two-stage rank relied on extracting a monochromatic length-r arithmetic progression in the stage-1 partition via the Hales–Jewett theorem [12]. Because Hales–Jewett numbers grow faster than primitive-recursively, that bound yields only $d - o(d)$ rank and collapses once M_1 grows as slowly as $\log \log d$; worse, Behrend colorings [2] avoid 3-term progressions with $M_1 = 2^{O(\sqrt{\log d})}$ classes, and random colorings cap monochromatic progressions at $O(\log d)$ even for $M_1 = 2$.

Wall 3. Aperiodic and Stone-shuffle blindness. As highlighted in [3, Problems 9.1 and 9.5], the cyclic shift rot_c couples only pairs (i, j) and (i', j') on the same anti-diagonal $i + j = i' + j'$ (when rows and shifts vary together across class-uniform collisions). Moreover, many shared-decoder invariants treat matrix transpose $\tau(X)_{i,j} = X_{j,i}$ and Stone's perfect shuffle/unshuffle permutation [15] on an equal footing, even though Stone's shuffle and unshuffle are computable in $O(L)$ steps on multi-tape Turing machines (3 work tapes for shuffle and 2 work tapes for unshuffle, without hardcoded d or b) in $O(1)$ passes!

Prior barrier / problem	Kernel-verified resolution in this paper	Theorem / Lean module
Wall 1 & [3, Problem 9.4(b)]: arbitrary sets reach rate $1 - o(1)$ at $\Delta = d^2$; spread windows $\Delta \geq 3$, $M = o(d/\Delta)$, and $\mathcal{E} \neq \emptyset$ open at rate 1.	Alphabet-independent capacity $\log_{\lvert\Sigma\rvert}\lvert C\rvert \leq d^2 - (d^2 - \lvert\mathcal{E}\rvert - dM\Delta)\frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}}$; Lean-verified rate < 1 for all $\Delta \geq 1$ and strict deficit for $M\Delta = o(d)$.	Theorems 3.4 and 3.7 and Corollary 3.5 (`BoundedAritySetCapacityProofs.lean`)
Wall 2: two-stage linear Hales–Jewett collapse at $M_1 \approx \log\log d$ ([9, Section 6]).	*Theorem 2A* (any field F): affine k-cubes give exact-factorization cutoff $d \leq \max(8M_1M_2^2, (2M_1)^{2^{M_2}})$ ($M_1 = o(d^{1/2^{M_2}})$ for fixed M_2, covering Behrend sets). *Theorem 2B* ($(d : F) \neq 0$): sumset annihilating polynomials + KST give $\Omega(bd(\log d)^{1/(2M_2-1)})$ rank deficit ($\Omega(bd\log d)$ at $M_2 = 1$).	Theorems 4.4 and 5.6 and Corollary 5.7 (`AffineCubeCompositionProofs.lean`, `SumsetRankDeficitProofs.lean`)
Wall 3 & [3, Problems 9.1, 9.5]: rot-invariance & Stone-shuffle blindness.	Anti-diagonal tensor untwisting $F^{d\times d} \cong F^d \otimes_F F^d + 5/8$ generic mask floor; Operator Schmidt Rank $\mathrm{rk}_\otimes(\tau) = d^2$ vs. $\mathrm{rk}_\otimes(S) = 4$, forcing $t \geq \log_2 d$ passes on shuffle/row/column pipelines.	Theorems 6.1 and 6.5 (`OperatorDualityProofs.lean`)
[3, Remark 5.6]: does $\sum_{\ell=1}^3 A_\ell X B_\ell = X^\top$ obey $\dim S \leq d(d+1)/2$?	**No** over every field: Kronecker family K_{3m} achieves $4m(m+1)$ (exceeding $\frac{3m(3m+1)}{2}$ for $m \in \{1,2,3,4\}$), plus $11 > 10$ at $d = 4$ and $29 > 28$ at $d = 7$ ($d \in \{3,4,6,7,9,12\}$).	Theorems 7.1 and 7.2 (`SwapCodeAnswersProofs.lean`)
[3, Problem 9.3]: multi-scale dyadic masks on both sides of X.	Interval bound $\frac{d^2 + d\min(\beta_A,\beta_B)}{2}$ with $\beta_\bullet = \max(1, 2\sum_j M_{\bullet,j})$ ($\frac{1}{2} + o(1)$ for sublinear profiles); Walsh boundary dichotomy (2^{j+1} boundaries vs. ≤ 2).	Theorems 7.3 and 7.4 (`SwapCodeAnswersProofs.lean`)
[3, Problem 9.4]: native TM transition-table sweep extraction.	Symbolic `CausalConeTree` & `SlotConeTree` extraction from $M.\delta$ on frozen trace cuts + conditional reductions of (T).	Theorems 8.2 to 8.4 (`MachineSweepConsequences.lean`)

Table 1: Summary of how the main results address the walls and open problems of [3, 9].

1.3 Overview of results and how they reshape the attack on (T)

This paper shows that reuniting *bounded read arity* ($\Delta = O(1)$) with *shared decoders* (M classes), replacing arithmetic progressions by *affine Hilbert cubes* and *shifted sumsets*, and tracking *Operator Schmidt Rank* across the row/column tensor bipartition breaks through all three walls simultaneously, answers the concrete algebraic questions of [3, Remark 5.6 and Problem 9.3], and builds the native transition-table bridge of [3, Problem 9.4]. Table 1 summarizes the correspondence between the walls and open problems of [3, 9] and our kernel-verified theorems.

Let us highlight six aspects of these results and their exact scope:

1. **Bounded-arity arbitrary-set capacity (Section 3).** Theorem 3.4 proves that for *every* finite alphabet $|\Sigma| \geq 2$, *every* subset $C \subseteq \Sigma^{d\times d}$, and *every* M-class deterministic shared decoder of cell read arity $\Delta \geq 1$ accurate outside $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$,

$$\log_{|\Sigma|}|C| \;\leq\; d^2 \;-\; (d^2 - |\mathcal{E}| - dM\Delta)\, p_{\min}(\Delta), \qquad p_{\min}(\Delta) \;:=\; \frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}} = \Theta(1/\Delta).$$

   At $\Delta = 1$ with $\mathcal{E} = \emptyset$, [3, Proposition 6.21(b)] gives the written cycle-counting rate $\frac{1}{2} + \frac{M}{2d}$, and at $M = 1, \Delta \leq 2, \mathcal{E} = \emptyset$, [3, Proposition 6.22(b)] gives a written row-ranking bound of

rate $\frac{1}{2}+\frac{3}{2d}$. For every constant arity $\Delta \geq 3$ on spread relations ([3, Problem 9.4(b)]), as well as across sublinearly growing class counts $M = o(d/\Delta)$ and broken-output sets $\mathcal{E} \neq \emptyset$ at every $\Delta \geq 1$, Theorem 3.4 is the *first* nontrivial (< 1) capacity bound on arbitrary nonlinear sets—and the first Lean-verified bound < 1 at $\Delta \geq 2$, giving e.g. $\leq \frac{23}{27}+\frac{8}{27d}$ at $M = 1, \Delta = 2$ (Corollary 3.5) and $\leq \frac{229}{256}+o(1)$ at $\Delta = 3$. Crucially, the formula has *zero dependence on* $|\Sigma|$: even over the canonical word alphabet $|\Sigma_d| = 2^{\lceil \log_2 d \rceil} \geq d$, where reading a single symbol can reveal the shift index c, a fixed read footprint $Q_{p(c),r}$ of size $\Delta < d$ misses the self-referential coordinate $(c, r-c)$ on at least $d - M\Delta$ shifts c, forcing the $1 - \Theta(1/\Delta)$ ceiling without counting functions over a fixed alphabet. Whether the $M = 1, \Delta = O(1)$ ceiling for $\Delta \geq 3$ drops all the way to $\frac{1}{2}+o(1)$ (the question of [3, Problem 9.4(b)]) or admits nonlinear coding gain above $\frac{1}{2}$ remains open and is formalized as an exact logical dichotomy (Proposition 3.9).

2. **Affine Hilbert cubes and Ramsey-free two-stage cutoffs (Section 4).** Over an *arbitrary* field F, we replace arithmetic progressions by affine k-cubes $c_0+\{0,\delta_1\}+\cdots+\{0,\delta_k\} \subseteq \mathbb{Z}_d$ with distinct nonzero steps δ_r. Lemma 4.2 proves that an affine k-cube in one stage-1 class forces $(k+1)b$ independent linear constraints per stage-2 class on the common message W (not a naive 2^k-multiplier recurrence, which fails already at $k = 2$: at $d = 13, q = 2$, Example 4.5 builds an explicit 3-AP-free 2-cube $J = \{0,1,4,5\} \subseteq \mathbb{Z}_{13}$ attaining $\dim W = 9 > 8 = \lfloor (26+6)/4 \rfloor$ over every field with $(13 : F) \neq 0$, while obeying $3 \dim W \leq 30 \implies \dim W \leq 10$ over every field). Because affine k-cubes are bipartite box patterns extractable by iterated Cauchy–Schwarz (Lemma 4.3), Theorem 4.4 eliminates Hales–Jewett numbers and proves the explicit cutoff $d \leq \max(8M_1M_2^2,\ (2M_1)^{2^{M_2}})$, ruling out exact two-stage linear factorizations for M_1 up to $\Theta(d^{1/2^{M_2}})$ at fixed M_2 (including 3-AP-free Behrend partitions with $M_1 = 2^{O(\sqrt{\log d})}$) and M_2 up to $(1-\varepsilon)\log_2\log_2 d$ at fixed M_1. As an essential coded-lane control at the boundary $d = 4 = M_1M_2$, Example 4.6 shows that even when K_1 and K_2 reuse the *exact same* two-interval partition $p_1 = p_2 = [0,0,1,1]$ on $\mathbb{Z}_4$, exact full transposition $K_2 \circ K_1 = \tau$ holds for all lane widths $b \geq 1$ over every field F with $(5 : F) \neq 0$ (including scalar $b = 1$ over $\mathbb{F}_2$!), whereas uncoded entry-copy and bit-lane-copy factorizations are impossible for all $b \geq 1$ and $|\Sigma| \geq 2$.

3. **Shifted sumsets and polynomial-in-$\log d$ two-stage rank deficits (Section 5).** Over any field F with $(d : F) \neq 0$, we go qualitatively beyond common-message growth for *fixed* class counts M_1, M_2. On any shifted $(2M_2)$-fold monochromatic sumset $\widetilde{J}_0+\widetilde{B}_1+\cdots+\widetilde{B}_{2M_2-1}$ in one stage-1 class with $|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t$, and $|\widetilde{B}_t| = k+1$, an injective arbitrary-subset Vandermonde map over roots of unity (Lemma 5.2) combines with a stacked band pullback (Lemma 5.2) and $2M_2 - 1$ sparse annihilating polynomials of degree $\leq \operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)$ (Lemma 5.3) to force $2\operatorname{rank}(\tau - K_2K_1) \geq bk\big(d - \sum_t \operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)\big)$ (Theorem 5.4). Combining this with iterated Kővári–Sós–Turán sumset extraction (Lemma 5.5) proves Theorem 5.6: for all constants $M_1, M_2 \geq 1$, every exact or approximate two-stage linear composition over $(d : F) \neq 0$ has error rank $\operatorname{rank}(\tau - K_2K_1) = \Omega\big(b\,d\,(\log d)^{1/(2M_2-1)}\big)$, and $\Omega(b\,d\log d)$ when $M_2 = 1$ (Corollary 5.7), compared to the sub-iterated-logarithmic $o(d)$ bound of [9, Theorem 6.6]. Example 5.8 proves over $\mathbb{C}$ at $d = 12$ why true stage-1 shift invariance cannot be relaxed to an abstract 1-class band subspace.

4. **Anti-diagonal untwisting and Operator Schmidt Rank duality (Section 6).** Conjugating by the anti-diagonal transform $Z_{c,a} = X_{c,\,a-c}$ intertwines rot_t with the pure right-factor shift $I_d \otimes \sigma^t$ on $F^d \otimes_F F^d$, decoupling shift-collision constraints into d independent anti-diagonal slices (Theorem 6.1), packaging the 9/16 and 5/8 random-mask certificate floors of [9, Appendix C], and realizing the single-shift 5/8 floor via generic tensor subspaces (Proposition 6.2).

Across the row/column tensor cut $\mathrm{End}_F(F^d \otimes F^d) \cong \mathrm{End}_F(F^d) \otimes \mathrm{End}_F(F^d)$, Operator Schmidt Rank cleanly separates matrix transpose ($\mathrm{rk}_\otimes(\tau) = d^2$, Theorem 6.4) from Stone's perfect shuffle ($\mathrm{rk}_\otimes(S) = 4$ for even $d = 2^m$, Theorem 6.4), giving a $t \geq \log_2 d$ pass lower bound on pipelines of row-local, column-local, and Stone-shuffle layers (Theorem 6.5) while certifying $O(L)$-time `Transpose.TM` implementations of 3-tape shuffle, 2-tape unshuffle, and 2-tape row shear (Proposition 6.6). Two controls (Propositions 6.7 and 6.8) delineate the exact scope of $\mathrm{rk}_\otimes$ on row-diagonal gauges and nonlinear one-time-pad encodings.

5. **Answers to Remark 5.6 and Problem 9.3 of Paper 17 (Section 7).** We resolve [3, Remark 5.6] in the negative over *every* field F: three-term swap subspaces satisfying $A_1XB_1 + A_2XB_2 + A_3XB_3 = X^\top$ attain exact dimension $4m(m+1)$ at $d = 3m$ for all $m \geq 1$ (strictly exceeding $\frac{3m(3m+1)}{2}$ for $m \in \{1,2,3,4\}$, i.e., $d \in \{3,6,9,12\}$), plus $11 > 10$ at $d = 4$ and $29 > 28$ at $d = 7$, exceeding $d(d+1)/2$ across $d \in \{3,4,6,7,9,12\}$ (Theorem 7.2), with $d = 5$ (15 vs. 15) isolated as a crisp open threshold. We also resolve the literal dyadic-interval formulation of [3, Problem 9.3], proving the upper bound $(d^2 + d\min(\beta_A, \beta_B))/2$ with $\beta_\bullet = \max(1, 2\sum_j M_{\bullet,j})$ and full d^2 attainment at the complete dyadic profile (Theorem 7.3), and establish the structural boundary dichotomy separating contiguous dyadic intervals (≤ 2 cyclic boundaries) from periodic Rademacher/Walsh masks (2^{j+1} cyclic boundaries, singleton-atom intersection, yet $\leq (m+1)2^m$ single-mask rot-span; Theorem 7.4).

6. **Native transition-table sweep extraction on frozen cuts (Section 8).** Addressing [3, Problem 9.4], we prove that on any frozen trace/cut fibre of a native `Transpose.TM` ($1 \leq M.k$, extendable from $M.k = 0$ via dummy-tape padding Lemma 2.2) whose cut-free causal paths have length $\leq D$, the machine's transition function $M.\delta$ evaluates explicit depth-D symbolic `CausalConeTree`s of cell support $\leq b(M.k+1)^D$. Periodic relative tree templates induce a genuine `IsBoundedAritySetCode` either by absorbing cut-touching outputs into $\mathcal{E}$ (Theorem 8.2) or—with *zero* extra broken cells—by refining M_0 via L_{cut} frozen cut-slot labels per column into $M_{\mathrm{cls}} = M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ classes (Theorem 8.3). We conclude in Section 8.3 with conditional theorems showing precisely which remaining trace-extraction, half-barrier, and macroscopic rank hypotheses would close (T).

1.4 Formalization status and conventions

Every numbered theorem, lemma, proposition, corollary, and example in this paper is completely formalized and kernel-verified in Lean 4 (version `v4.34.1` with Mathlib [16]) inside the `BoundedArityDecoders` library, depending solely on Lean's three standard foundational axioms (`propext`, `Classical.choice`, and `Quot.sound`), with zero `sorry`, `admit`, `native_decide`, or custom axioms. Each numbered result carries a `Lean`: header naming its formal declarations and source module, followed by a self-contained mathematical proof outline faithful to the formal proof. Results quoted in prose from earlier papers of the series record their exact manuscript locator (noting explicitly whenever an upstream statement is written-only or conditional); where we formalize the separated storage-code amplification benchmark from [10] in Proposition 2.5(a), its formal statement retains the explicit seed hypothesis `HasOddGirthLinearSeed` (matching `TapeStorageRates.OddGirthLinearSeed` of [10, Definition 5.1]). Question (T) itself remains open; all implications toward (T) in Section 8 are stated with their exact formal hypotheses.

2 Models, shared decoders, and prior-spine benchmarks

2.1 Matrix states, cyclic shifts, and bounded-arity shared set codes

For $d \geq 1$, write $\mathbb{Z}_d = \mathbb{Z}/d\mathbb{Z}$ (formalized as `ZModd`). Given an entry alphabet Σ, a $d \times d$ matrix state is a map $X : \mathbb{Z}_d \to \mathbb{Z}_d \to \Sigma$, written $X \in \Sigma^{d\times d}$ (`SharedDecoders.State`). Its transpose $\tau(X) = X^\top$ (`SharedDecoders.swap`) and column-cyclic shift $\mathrm{rot}_c(X)$ for $c \in \mathbb{Z}_d$ (`SharedDecoders.rot`) are defined by

$$\tau(X)_{i,j} = X_{j,i}, \qquad \mathrm{rot}_c(X)_{i,j} = X_{i,\,j+c} \qquad (i, j \in \mathbb{Z}_d).$$

Thus column 0 of $\mathrm{rot}_c(X)$ is column c of X, whereas column c of $\tau(X)$ is the transposed vector $r \mapsto X_{c,r}$ (row c of X).

Definition 2.1 (Syntactic read footprints and bounded-arity shared set codes)**.** Let $d \geq 1$, let $M \geq 0$ be a number of decoder classes, let $\Delta \geq 0$ be a read-arity bound, and let Σ be a finite alphabet.

(a) A family of class decoders $f : \{0, \ldots, M-1\} \to \Sigma^{d\times d} \to \Sigma^{\mathbb{Z}_d}$ has *syntactic cell support* $\leq \Delta$ with footprint table $Q : \{0, \ldots, M-1\} \to \mathbb{Z}_d \to \binom{\mathbb{Z}_d \times \mathbb{Z}_d}{\leq \Delta}$ (`SyntacticCellSupport`) if $|Q_{m,r}| \leq \Delta$ for every class m and output coordinate $r \in \mathbb{Z}_d$, and whenever $Y_1, Y_2 \in \Sigma^{d\times d}$ agree on every cell $q \in Q_{m,r}$, one has $f_m(Y_1)_r = f_m(Y_2)_r$. (Its whole-column counterpart reading at most Δ_c columns of Y is `SyntacticColumnSupport`.)

(b) Given a column-class assignment $p : \mathbb{Z}_d \to \{0, \ldots, M-1\}$, a broken-output set $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, and a subset $C \subseteq \Sigma^{d\times d}$, the data $(\Delta, p, \mathcal{E}, f, Q)$ form a *bounded-arity shared set code on* C (`IsBoundedAritySetCode`) if (f, Q) has syntactic cell support $\leq \Delta$ and

$$f_{p(c)}\big(\mathrm{rot}_c(X)\big)_r \;=\; X_{c,r} \qquad \text{for all } X \in C \text{ and all } (r, c) \notin \mathcal{E}.$$

When $\Sigma = V$ is an F-module (typically $V = F^b$ for b-bit or b-symbol lanes), `SharedDecoders.IsSharedCode` ([9, Definition 2.1]) restricts the source $C \leq \mathrm{Mat}_{d\times d}(V)$ to an F-linear submodule while leaving the class maps f_m arbitrary, whereas linear class-uniform rules (`SwapRigidity.IsClassUniformCode` and `SharedDecoders.LinearClassUniform`, [3, Definition 6.3]) further require each f_m to be F-linear, without a cell-arity cap unless explicitly imposed.

2.2 Native multi-tape Turing machines, canonical words, and frozen cuts

We use the frozen multi-tape Turing machine model `Transpose.TM` imported unchanged from the foundational Lean package of [5, Definition 2.1]: a machine M has state set $\mathrm{Fin}(M.q + 1)$, work-tape alphabet $\mathrm{Fin}(M.g + 1)$, $M.k$ read-write work tapes, one two-way read-only binary input tape of length $L = d^2 b$ where $b =$ `Transpose.entryBits`(d), and one one-way write-only binary output tape. The canonical word alphabet (`CanonicalWordAlphabet`) is $\Sigma_d := \{0,1\}^b$ (`Fin(Transpose.entryBitsd)->Bool`), of exact cardinality $|\Sigma_d| = 2^b \geq d$.

Lemma 2.2 (Canonical word width and dummy-tape padding)**.** Lean: `entryBits_eq_clog` [`NativeSweepInterfaceProofs.lean`]
Lean: `padDummyWorkTape_transposesWithin` [`MachineSweepConsequences.lean`]

For every $d \geq 2$, `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ *and* $d \leq |\Sigma_d| = 2^{\lceil \log_2 d \rceil} < 2d$*. Moreover, for every time bound $T : \mathbb{N} \to \mathbb{N}$ and every machine M satisfying* `Transpose.TransposesWithinMT`*, there exists a machine M^+ (*`padDummyWorkTape_transposesWithin`*) with $1 \leq M^+.k$ that also satisfies* `Transpose.TransposesWithin` M^+ T*.*

Outline of the formal proof. The identity `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ is proved from `Nat.clog` on binary logarithms, and $|\Sigma_d| = 2^b$ follows by cardinality of function types over $\mathrm{Fin}\, b$. For dummy-tape padding (`padDummyWorkTape_transposesWithin`), when $M.k \geq 1$ one sets $M^+ = M$; when $M.k = 0$, M^+ sets $k = 1$, $g = 0$, and $q = M.q$ with one stationary blank work tape whose transition ignores its read symbol, writes 0, and stays stationary, maintaining identical control states, input-head positions, and emitted output bits at every step. □

To interface `Transpose.TM` both with F-linear matrix operators and with nonlinear cut fibres, we record two exact bridges:

(a) *Frozen trace/cut fibres* (`IsFrozenTraceCutFibre`): given M, dimension d, horizon T, head trajectories $\mathrm{inPos} : \mathbb{N} \to \mathbb{Z}$ and $\mathrm{workPos} : \mathbb{N} \to \mathrm{Fin}(M.k) \to \mathbb{Z}$, output emission times $\mathrm{emitStep} : \mathrm{Fin}(L) \to \{0, \ldots, T-1\}$, and a set of cut steps $\Gamma \subseteq \{0, \ldots, T-1\}$, a subset $C \subseteq \Sigma_d^{d \times d}$ is a *frozen trace/cut fibre* if every $X \in C$ induces the head trajectories $(\mathrm{inPos}, \mathrm{workPos})$, emits output cell (r, c)'s ℓ-th bit $X_{c,r,\ell}$ at step $\mathrm{emitStep}($`outputBitIdx`$(d, r, c, \ell))$, and all $X, X' \in C$ have identical post-step control state and written work-tape symbol at the accessed cell of step u for every cut step $u \in \Gamma$.

(b) *Matrix-to-grid linear bridge*: for every semiring F and $d \geq 0$, `encodeTMMatrixToGrid` embeds each Boolean bit matrix $A \in$ `Transpose.Matrix` into `GridState` F d $b \cong \mathrm{Mat}_{d \times d}(F^b)$ ($b =$ `Transpose.entryBits`(d)) via $0/1 \in F$ on each bit lane.

Proposition 2.3 (Equivariance of the native matrix-to-grid bridge)**.** Lean: `tmMatrix_gridLinear_bridges` [`DiagnosisAndBridgesConsequences.lean`]

For every ring F, `encodeTMMatrixToGrid` *intertwines* `stoneShuffleTMMatrix` *with* `stoneShuffleGridLinear` *(for $d = 2^m$),* `stoneUnshuffleTMMatrix` *with* `stoneUnshuffleGridLinear` *(for $d = 2^m$, paired with its linear-time 2-work-tape machine* `stoneUnshuffleTM_twoTape_linear`*),* `rowShearTMMatrix` *with* `rowShearLinear` *(for $d \geq 1$), and* `Transpose.transpose` *with* `gridSwapLinear`*.*

Outline of the formal proof. Verified directly by unfolding the 0/1 grid embedding `encodeTMMatrixToGrid` and checking that cell/lane permutations commute with $\{0, 1\} \hookrightarrow F$. □

2.3 Formalized benchmarks from the Machine and Shared-Decoder Spines

Before proving our new bounds, we formalize inside `BoundedArityDecoders` the exact quantitative benchmarks from the Machine Spine that motivate reuniting $\Delta = O(1)$ with decoder sharing. Doing so keeps the distinctions between word-step cones, bit-step cones, and Paper 18's conditional adaptive interior-lane cones completely explicit.

Proposition 2.4 (Bounded-cone reductions across trace models)**.** Lean: `diagnosis_TCPAndAdaptiveConeReduction` [`DiagnosisAndBarriersProofs.lean`]
Lean: `tmWholeCellCutConeBound`, `wordStepTrace_TCPCellConeReduction`, `diagnosis_Paper18AdaptiveOriginalEntryPullback` [`DiagnosisAndBridgesConsequences.lean`]
Lean: `Diagnosis_TCPAndAdaptiveConeReductionSpec` [`DiagnosisAndBarriers.lean`]

(a) TCP(k) walk-DAG and word-step cone compression *([4, Proposition 2.4 and Theorem 7.5]): for every $k \geq 1$ and $\varepsilon > 0$, there exists a constant depth $D = D(k, \varepsilon) \geq 1$ such that every T-step k-head unit-speed walk admits a cut step set $\Gamma \subseteq \{0, \dots, T-1\}$ of size $|\Gamma| \leq \varepsilon T$ reducing all Γ-free causal paths to $\leq D$ vertices and all Γ-truncated causal cones to $\leq (k+1)^D$ steps; on a word-step frozen trace (*`WordStepFrozenTrace`*), $D =$ `wordStepTCPDepth`(k, ε) bounds every output cell's input-cell cone by $(k+1)^D$ (*`wordStepTrace_TCPCellConeReduction`*).*

(b) Bit-level `Transpose.TM` fibres and whole-cell cones*: for every $c_0 L$-step transposer M ($1 \leq M.k$) and $0 < \varepsilon < 1$, there exist $D \geq 1$ and D_0 such that for all $d \geq \max(D_0, 2)$, a nonlinear fibre $C \subseteq \Sigma_d^{d \times d}$ of size $|C| \geq 2^{(1-\varepsilon)L}$ determines every output bit outside $\leq \varepsilon L$ broken bits from at most $(M.k+1)^D$ input cells; on any frozen trace/cut fibre of path length $\leq D$, the b bits of an output cell (r, c) jointly read at most $b(M.k+1)^D$ input cells (*`tmWholeCellCutConeBound`*).*

(c) Conditional adaptive interior-lane cut pullback *([11, Lemma H.2, Theorem H.5, and Corollary H.6]): alongside the per-entry window-certificate Markov tail, for any adaptive transposer (*$24s \leq b$*) on a common simulator-trace fibre S,* given *an original vertex cut B whose B-free paths have at most D vertices, B bounds original-entry capacity with whole-entry cone $\leq b(M.k+3)^D$ and transports to a simulator guard Γ of size $\leq G_s|B|$ outside which every retained interior bit has simulator input-bit cone $\leq 2(6s+1)(M.k+3)^D$ (*`diagnosis_Paper18AdaptiveOriginalEntryPullback`*).*

Outline of the formal proof. Part (a) invokes `BoundedStackDepth.genie` ([4]) on the k-head walk DAG and inducts over the $(k+1)$-ary dependency tree. Part (b) pigeonholes the aligned trace/cut advice on $c_0 L$-step runs and takes the b-fold union of bit-level cones. Part (c) bridges to `AdaptiveTransfer.TransferEntry.uniform_entry_success_tail`, `AdaptiveTransfer.TransferEntry.native_original_entry_capacity`, and `AdaptiveTransfer.TransferCones.native_cone_transport_interior`. □

Proposition 2.5 (Why cones without sharing stall: separated storage codes and barrel-shift networks)**.** Lean: `diagnosis_AbstractSeparatedHighRateCode`, `diagnosis_ShiftAndRotationCodingGainBarrier` [`DiagnosisAndBarriersProofs.lean`]
Lean: `diagnosis2_HighRateTendstoOne`, `diagnosis3_MultiBitBlockEdgeShiftBarrier` [`DiagnosisAndBridgesConsequences.lean`]
Lean: `Diagnosis_AbstractSeparatedHighRateCodeSpec`, `Diagnosis_ShiftAndRotationCodingGainBarrierSpec` [`DiagnosisAndBarriers.lean`]

(a) High-rate binary storage codes under paired-distance separation *([6, Theorem 7.1 and Corollary 8.2]; [10, Definition 5.1, Theorem 5.5, and Theorem 6.1]): the Goren–Oort numerical parameters $(p, k) = (127, 91)$ exceed half rate both over $\mathbb{F}_{127^2}$ ($8100/16130 > 1/2$) and after 14-bit binary blow-up (> 0.5013); moreover, given any finite field K of odd characteristic ($(2 : K) \neq 0$) and odd-girth linear seed* `HasOddGirthLinearSeed` K D_{seed} ρ *with $\frac{1}{2} < \rho < 1$, there exist $C_{\mathrm{code}} > 0$, $0 < \alpha < \frac{1}{2}$, and Δ_0 such that every degree cap $\Delta \geq \Delta_0$ and paired-distance girth threshold G admit a nonempty (Δ, G)-separated binary storage code $C \subseteq \{0,1\}^n$ of rate $\frac{\log_2 |C|}{n} \geq 1 - C_{\mathrm{code}} \Delta^{-\alpha} \to 1$ as $\Delta \to \infty$ (*`diagnosis2_HighRateTendstoOne`*).*

(b) Barrel-shift complete-exchange coding-gain barrier *([5, Propositions 11.2 and 11.3]): any size-s fan-in-2 acyclic Boolean barrel-shift circuit on $n \geq N_0$ bits (*`HasBarrelShiftCircuit`*) embeds into a max-degree-4 graph on $V \leq C_{\mathrm{net}}(s+n)$ vertices carrying an n-round causal complete-exchange code (*`HasCausalCompleteExchangeCode`*) whose coding-to-routing gain against any multicommodity routing rate $\lambda_{\mathrm{route}} > 0$ is at least $c_{\mathrm{gain}}\, n \log_2 n/(s+n)$ (and $\geq c_{\mathrm{gain}}\, n \log_2 n/(ws_{\mathrm{blk}}+n)$ on s_{blk} w-bit block edges,* `diagnosis3_MultiBitBlockEdgeShiftBarrier`*), while every $n \geq 2$ admits an explicit barrel shifter of size $s \leq C_{\mathrm{sh}} n \log_2 n$.*

Outline of the formal proof. Part (a) verifies the $(127, 91)$ arithmetic inequalities, amplifies the odd-characteristic linear seed by recursive odd-majority tensor products on syndrome fibres, and takes $\Delta \to \infty$. Part (b) converts each fan-in-2 gate into a bounded-degree DAG gadget serving all n cyclic shift demands in n rounds, bounds pairwise source-to-sink walk lengths via the $n!$ shift permutations ($\sum_{i,j} \mathrm{distBound}(i,j) \geq c_{\mathrm{dist}} n^2 \log_2 n$), and builds the $O(n \log n)$ barrel shifter from $\lceil \log_2 n \rceil$ power-of-two multiplexor layers. □

Part I
Bounded-arity capacity and Ramsey-free composition

3 Bounded-arity shared decoders on arbitrary sets

3.1 Shifted footprints, self-reference, and one-hop acyclic recovery

In [9, Appendix A, Theorem A.1 and Corollary A.2], an arbitrary nonlinear subset $C \subseteq \Sigma^{d \times d}$ validated by a single shared decoder ($M = 1$) of unrestricted read arity ($\Delta = d^2$) reaches rate $1 - \frac{\log_{|\Sigma|} d}{d^2} \to 1$ via cyclic-orbit markers. In [3, Section 6.7, Propositions 6.21–6.23, and Problem 9.4(b)], acyclic-subgraph index-coding bounds ([1, Theorem 3]) were used to give written proofs for bounded offsets ([3, Proposition 6.21(a)]), window one ($\Delta = 1$, [3, Proposition 6.21(b)]), exact 1-class window two ($M = 1, \Delta \leq 2, \mathcal{E} = \emptyset$, [3, Proposition 6.22(b)]), and bounded-value offsets or mutual helpers ([3, Proposition 6.22(c)–(d) and Proposition 6.23]), leaving windows $\Delta \geq 3$ with spread rows and offsets, growing class counts $M = o(d/\Delta)$, and broken outputs $\mathcal{E} \neq \emptyset$ open at rate 1. Over the canonical word alphabet $\Sigma_d = \{0,1\}^{\lceil \log_2 d \rceil}$ ($|\Sigma_d| \geq d$), a single entry $(\mathrm{rot}_c X)_{0,0} = c$ can already reveal the shift index c to a nonlinear rule, blocking any naive reduction to fixed-alphabet function counting.

The mechanism that defeats arbitrary nonlinear coding at every bounded arity $\Delta \geq 1$ is geometric rather than alphabet-dependent. Suppose $(\Delta, p, \mathcal{E}, f, Q)$ is a bounded-arity shared set code on $C \subseteq \Sigma^{d \times d}$ (Definition 2.1). When class $m = p(c)$ computes transposed output (r, c)—whose required value on $X \in C$ is matrix entry $X_{c,r}$ whenever $(r, c) \notin \mathcal{E}$—it evaluates $f_{p(c)}(\mathrm{rot}_c(X))_r$, which depends only on the shifted read footprint (`shiftedFootprint`)

$$F_{c,r} \;:=\; \{\, (i,\, c+u) \in \mathbb{Z}_d \times \mathbb{Z}_d \;:\; (i,u) \in Q_{p(c),\,r} \,\}$$

of cardinality $|F_{c,r}| \leq \Delta$. Crucially, (c, r) lies in its own footprint $F_{c,r}$ if and only if $(c,\, r-c) \in Q_{p(c),r}$. For each fixed class m and output coordinate r, the map $c \mapsto (c, r-c)$ is injective on $\mathbb{Z}_d$, so a fixed set $Q_{m,r}$ of size at most Δ can contain $(c, r-c)$ for at most Δ columns c!

Lemma 3.1 (Self-referential characterization and clean-entry count)**.** Lean: `selfReferentialAndCleanCount` [`BoundedAritySetCapacityProofs.lean`]
Lean: `SelfReferentialAndCleanCountSpec` [`BoundedAritySetCapacity.lean`]
Let $d \geq 1$, $M, \Delta \in \mathbb{N}$, $p : \mathbb{Z}_d \to \operatorname{Fin} M$, $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, and $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$.

(a) For all $c, r \in \mathbb{Z}_d$, $(c, r) \in F_{c,r}$ holds if and only if $(c, r - c) \in Q_{p(c),r}$.

(b) Define the set of clean matrix entries *(`cleanEntries`) by*

$$\mathcal{G} \;:=\; \{\, (c, r) \in \mathbb{Z}_d \times \mathbb{Z}_d \;:\; (r, c) \notin \mathcal{E} \text{ and } (c, r) \notin F_{c,r} \,\}.$$

If $|Q_{m,r}| \leq \Delta$ for all (m, r), then every $(c, r) \in \mathcal{G}$ satisfies $|F_{c,r}| \leq \Delta$, and

$$|\mathcal{G}| \;\geq\; d^2 - |\mathcal{E}| - dM\Delta.$$

Outline of the formal proof. Part (a) follows immediately from $(i, c + u) = (c, r) \iff (i, u) = (c, r - c)$. For part (b), summing over $r \in \mathbb{Z}_d$ and $m \in \operatorname{Fin} M$, the set of columns c with $p(c) = m$ and $(c, r - c) \in Q_{m,r}$ injects into $Q_{m,r}$ via $c \mapsto (c, r - c)$, contributing at most Δ self-referential pairs per (r, m) and hence at most $dM\Delta$ overall; removing those and the $|\mathcal{E}|$ transposed broken positions (c, r) with $(r, c) \in \mathcal{E}$ leaves at least $d^2 - |\mathcal{E}| - dM\Delta$ clean entries, each with $|F_{c,r}| \leq |Q_{p(c),r}| \leq \Delta$. □

For any seed set $\Omega \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$ of revealed matrix coordinates, define the one-hop recoverable set (`recoverableCells`)

$$\operatorname{Rec}(\Omega) \;:=\; \{\, (c, r) \in \mathbb{Z}_d \times \mathbb{Z}_d \;:\; (c, r) \notin \Omega,\ (r, c) \notin \mathcal{E},\ \text{and } F_{c,r} \subseteq \Omega \,\}.$$

Lemma 3.2 (One-hop acyclic determination)**.** Lean: `oneHopRecoverableInjective` [`BoundedAritySetCapacityProofs.lean`]
Lean: `OneHopRecoverableInjectiveSpec` [`BoundedAritySetCapacity.lean`]
If $(\Delta, p, \mathcal{E}, f, Q)$ is a bounded-arity shared set code on $C \subseteq \Sigma^{d \times d}$, then for every $\Omega \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, restricting codewords $X \in C$ to $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus \operatorname{Rec}(\Omega)$ is injective, and hence

$$|C| \;\leq\; |\Sigma|^{\,d^2 - |\operatorname{Rec}(\Omega)|}.$$

Outline of the formal proof. Suppose $X_1, X_2 \in C$ agree on $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus \operatorname{Rec}(\Omega)$. Since $\Omega \cap \operatorname{Rec}(\Omega) = \emptyset$, they agree on all of Ω. For every $(c, r) \in \operatorname{Rec}(\Omega)$, $F_{c,r} \subseteq \Omega$ means $\operatorname{rot}_c(X_1)$ and $\operatorname{rot}_c(X_2)$ agree on $Q_{p(c),r}$; because $(r, c) \notin \mathcal{E}$, syntactic cell support gives $X_1(c, r) = f_{p(c)}(\operatorname{rot}_c X_1)_r = f_{p(c)}(\operatorname{rot}_c X_2)_r = X_2(c, r)$. Injection into functions $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus \operatorname{Rec}(\Omega) \to \Sigma$ yields the cardinality bound. □

3.2 Theorem 1: alphabet-independent capacity of bounded-arity set codes

To maximize $|\operatorname{Rec}(\Omega)|$ simultaneously across arbitrary footprints of size $\leq \Delta$, we sample $\Omega \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$ by including each cell independently with probability $\theta = \frac{\Delta}{\Delta+1}$ (`bernoulliSubsetWeight`). Write

$$p_{\min}(\Delta) \;:=\; \frac{\Delta^{\Delta}}{(\Delta + 1)^{\Delta+1}} \qquad (\texttt{arityGainFactor}).$$

Lemma 3.3 (First-moment recoverable subset and $\Theta(1/\Delta)$ scaling)**.** Lean: `existsRecoverableSubsetBound` [BoundedAritySetCapacityProofs.lean]
Lean: `pMin_two_sided_theta_inv_arity` [SetCapacityConsequences.lean]
Lean: `ExistsRecoverableSubsetBoundSpec` [BoundedAritySetCapacity.lean]

Let $d \geq 1$, $M \geq 1$, $\Delta \geq 1$, $p : \mathbb{Z}_d \to \operatorname{Fin} M$, $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, and $|Q_{m,r}| \leq \Delta$ for all (m, r).

(a) Under product Bernoulli$(\frac{\Delta}{\Delta+1})$ *measure on $\Omega \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, every clean entry $(c, r) \in \mathcal{G}$ falls into* $\mathrm{Rec}(\Omega)$ *with probability $(\frac{\Delta}{\Delta+1})^{|F_{c,r}|} \frac{1}{\Delta+1} \geq p_{\min}(\Delta)$, and there exists a deterministic subset $\Omega_* \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$ satisfying*

$$|\mathrm{Rec}(\Omega_*)| \;\geq\; |\mathcal{G}|\, p_{\min}(\Delta) \;\geq\; \left(d^2 - |\mathcal{E}| - dM\Delta\right) p_{\min}(\Delta).$$

(b) For every integer $\Delta \geq 1$,

$$\frac{1}{e(\Delta+1)} \;<\; p_{\min}(\Delta) \;\leq\; \frac{1}{2(\Delta+1)} \;\leq\; \frac{1}{4}.$$

Outline of the formal proof. For $(c, r) \in \mathcal{G}$, $(c, r) \notin F_{c,r}$, so the events $\{(c, r) \notin \Omega\}$ and $\{F_{c,r} \subseteq \Omega\}$ depend on disjoint coordinate sets and have product probability $(1-\theta)\theta^{|F_{c,r}|} \geq (1-\theta)\theta^{\Delta} = p_{\min}(\Delta)$ since $|F_{c,r}| \leq \Delta$ and $\theta = \frac{\Delta}{\Delta+1} \in (0, 1)$. Linearity of expectation over finite subsets of $\mathbb{Z}_d \times \mathbb{Z}_d$ yields Ω_* achieving at least the expectation. In part (b), $(1 + \frac{1}{\Delta})^{\Delta} < e$ gives the strict lower bound, while binomial induction proves $2\Delta^{\Delta} \leq (\Delta+1)^{\Delta}$, whence $p_{\min}(\Delta) \leq \frac{1}{2(\Delta+1)} \leq \frac{1}{4}$. □

Combining Lemmas 3.1 to 3.3 yields our first main theorem, in both exponential and unconditional logarithmic forms (including empty or nonempty C and the growing canonical word alphabet Σ_d).

Theorem 3.4 (Bounded-arity arbitrary-set shared-decoder capacity)**.** Lean: `theorem1_BoundedAritySetCapacity` [BoundedAritySetCapacityProofs.lean]
Lean: `theorem1_BoundedAritySetCapacity_unconditional`, `theorem1_CanonicalWordSetCapacity`, `canonicalWordAlphabet_card_and_ge_two` [SetCapacityConsequences.lean]
Lean: `Theorem1_BoundedAritySetCapacitySpec` [BoundedAritySetCapacity.lean]

Let $d \geq 1$, $M \in \mathbb{N}$, $\Delta \geq 1$, and let Σ be any finite alphabet with $|\Sigma| \geq 2$. For every column-class map $p : \mathbb{Z}_d \to \operatorname{Fin} M$, broken-output set $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, footprint-bounded decoders (f, Q), and arbitrary (possibly nonlinear, empty or nonempty) subset $C \subseteq \Sigma^{d \times d}$ satisfying `IsBoundedAritySetCode` *$(\Delta, p, \mathcal{E}, f, Q, C)$,*

$$|C| \;\leq\; |\Sigma|^{\,d^2 - (d^2 - |\mathcal{E}| - dM\Delta)\, p_{\min}(\Delta)} \quad \text{and} \quad \frac{\log|C|}{\log|\Sigma|} \;\leq\; d^2 - (d^2 - |\mathcal{E}| - dM\Delta)\frac{\Delta^{\Delta}}{(\Delta+1)^{\Delta+1}}.$$

In particular, both bounds hold unconditionally on the canonical word alphabet $\Sigma_d = \{0,1\}^{\lceil \log_2 d \rceil}$ ($|\Sigma_d| = 2^{\lceil \log_2 d \rceil}$ for $d \geq 2$).

Outline of the formal proof. When $C \neq \emptyset$ (which also forces $M \geq 1$ via $p(0) \in \operatorname{Fin} M$), choosing Ω_* from Lemma 3.3(a) inside Lemma 3.2 gives $|C| \leq |\Sigma|^{d^2 - |\mathrm{Rec}(\Omega_*)|} \leq |\Sigma|^{d^2 - (d^2 - |\mathcal{E}| - dM\Delta) p_{\min}(\Delta)}$; taking real logarithms and dividing by $\log|\Sigma| > 0$ proves the logarithmic inequality. When $C = \emptyset$, $\operatorname{Nat.card} C = 0$ makes the left-hand side 0 under Lean's convention $\log 0 = 0$, while $p_{\min}(\Delta) \leq 1/4$ (Lemma 3.3(b)) ensures the right-hand side is at least $\frac{3}{4}d^2 \geq 0$. □

Corollary 3.5 (Finite-d one-class ceilings at $\Delta = 1$ and $\Delta = 2$)**.** Lean: `corollary_OneClassArityOneThreeQuarters`, `corollary_OneClassArityTwoTwentyThreeTwentySevenths` [`SetCapacityConsequences.lean`]

Let $d \geq 1$, $|\Sigma| \geq 2$, $p : \mathbb{Z}_d \to \operatorname{Fin} 1$, and $\mathcal{E} = \emptyset$. For every subset $C \subseteq \Sigma^{d\times d}$ validated by a 1-class shared decoder of cell arity Δ:

(a) if $\Delta = 1$, then $\dfrac{\log|C|}{d^2 \log|\Sigma|} \leq \dfrac{3}{4} + \dfrac{1}{4d}$;

(b) if $\Delta = 2$, then $\dfrac{\log|C|}{d^2 \log|\Sigma|} \leq \dfrac{23}{27} + \dfrac{8}{27d}$.

Outline of the formal proof. Substitute $M = 1$, $|\mathcal{E}| = 0$, and $p_{\min}(1) = 1/4$ or $p_{\min}(2) = 4/27$ into Theorem 3.4 and divide by $d^2 > 0$. □

Remark 3.6 (Comparison with Section 6.7 of Paper 17)**.** At $\mathcal{E} = \emptyset$, the written (unformalized) bounds of [3, Proposition 6.21(b) and Proposition 6.22(b)] give rate $\frac{1}{2} + \frac{M}{2d}$ at $\Delta = 1$ (by cycle counting) and rate $\frac{1}{2} + \frac{3}{2d}$ at $M = 1, \Delta \leq 2$ (by a multigraph vertex-ranking argument on row indices), whereas one-hop Bernoulli sampling gives the Lean-verified bounds $\frac{3}{4} + \frac{M}{4d}$ at $\Delta = 1$ and $\frac{23}{27} + \frac{8}{27d}$ at $M = 1, \Delta = 2$ (along with handling $M = o(d)$ and $\mathcal{E} \neq \emptyset$). Once $\Delta \geq 3$ and both rows and offsets are spread, however, row rankings no longer force half-space acyclic sets, and [3, Section 6.7 and Problem 9.4(b)] had no bound below rate 1. Theorem 3.4 supplies an unconditional, Lean-verified strictly-below-1 barrier at *every* finite arity $\Delta \geq 1$ (e.g. $\leq \frac{229}{256} + \frac{81}{256d}$ at $M = 1, \Delta = 3$), uniformly over fixed and growing alphabets, sublinear class counts, and broken-output sets.

Theorem 3.7 (Positive rate deficit for sublinear classes and broken-output fractions)**.** Lean: `growingClassesPositiveDeficit` [`BoundedAritySetCapacityProofs.lean`]
Lean: `growingClasses_strict_inv_exp_deficit`, `asymptotic_SublinearClassesBrokenFractionRefutesRateOne` [`SetCapacityConsequences.lean`]
Lean: `GrowingClassesPositiveDeficitSpec` [`BoundedAritySetCapacity.lean`]

(a) Let $d \geq 1$, $\Delta \geq 1$, $|\Sigma| \geq 2$, and $\delta, \eta \geq 0$ with $\delta + \eta < 1$. Whenever $|\mathcal{E}| \leq \delta d^2$ and $M \leq \eta d/\Delta$, every $(\Delta, p, \mathcal{E})$-bounded-arity set code $C \subseteq \Sigma^{d\times d}$ satisfies the strict inequality

$$\frac{\log|C|}{d^2 \log|\Sigma|} < 1 - \frac{1 - \delta - \eta}{e(\Delta + 1)}.$$

(b) Asymptotically, for any fixed arity $\Delta \geq 1$, unbroken output fraction $\delta_{\text{ok}} \in (0, 1]$ ($|\mathcal{E}_d| \leq (1 - \delta_{\text{ok}})d^2$), and class count $M_d = o(d/\Delta)$, the normalized rate is eventually bounded by $1 - \frac{\delta_{\text{ok}}}{2} p_{\min}(\Delta) < 1$, refuting convergence of the rate to 1.

Outline of the formal proof. Under $|\mathcal{E}| \leq \delta d^2$ and $dM\Delta \leq \eta d^2$, the clean fraction satisfies $(d^2 - |\mathcal{E}| - dM\Delta)/d^2 \geq 1 - \delta - \eta > 0$; multiplying by $p_{\min}(\Delta) > \frac{1}{e(\Delta+1)}$ (Lemma 3.3(b)) proves (a) for nonempty C, and the empty-set check uses $\frac{1-\delta-\eta}{e(\Delta+1)} < \frac{1}{4}$. Part (b) chooses d large enough that $M_d\Delta/d \leq \delta_{\text{ok}}/2$ and applies Theorem 3.4. □

3.3 Boundary sharpness at $\Delta \geq d$ and the $M = 1, \Delta = O(1)$ dichotomy

Next we verify that the $dM\Delta$ subtraction in Theorem 3.4 marks a genuine phase transition at $\Delta = \Theta(d)$ rather than an artifact of the proof, and we formalize the exact logical relationship between the remaining $M = 1, \Delta = O(1)$ rate questions.

Proposition 3.8 (Sharpness at $\Delta \geq d$ and exact all-dimension Wall 1 asymptotic)**.** Lean: `boundaryControl_DeficitVanishesAtDeltaEqD` [`BoundedAritySetCapacityProofs.lean`]
Lean: `wall1_UnrestrictedArityRateLowerBoundAndTendstoOne`, `wall1_ExactLogarithmicAsymptoticAllDimensions` [`SetCapacityConsequences.lean`]
Lean: `BoundaryControl_DeficitVanishesAtDeltaEqDSpec` [`BoundedAritySetCapacity.lean`]

(a) Threshold at $\Delta = d$ and row-0 beacon codes*: at $M = 1, \mathcal{E} = \emptyset$, the clean-entry term $d^2 - d\Delta$ vanishes at $\Delta = d$. Moreover, for $d \geq 2$ and* every *finite alphabet $|\Sigma| \geq 2$, there exist an explicit* 1*-class zero-error shared set code at exact cell support $\Delta = d$ of size $\lfloor |\Sigma|^d/d \rfloor^d$, and an explicit* 1*-class zero-error shared set code using column support d and cell support at most $2d-1$ of size $|\Sigma|^{d(d-1)}$ (rate $1 - 1/d \to 1$).*

(b) Unrestricted-arity orbit code for all dimensions*: at $M = 1, \mathcal{E} = \emptyset, \Delta = d^2$, for every fixed $|\Sigma| \geq 2$, choosing one representative per free* rot*-orbit achieves rate $\geq 1 - \frac{\log_{|\Sigma|}(2d)}{d^2}$ for all $d \geq 2$ (and $= 1 - \frac{\log_{|\Sigma|} d}{d^2}$ for prime d), with exact all-dimension remainder*

$$\lim_{d\to\infty} \Big(\log_{|\Sigma|} |C_d| - \big(d^2 - \log_{|\Sigma|} d\big) \Big) = 0.$$

Outline of the formal proof. In (a), at exact cell arity $\Delta = d$, partitioning Σ^d into d-tuples of cyclic-shift representatives gives $\lfloor |\Sigma|^d/d \rfloor$ choices per row and size $\lfloor |\Sigma|^d/d \rfloor^d$; for the $(2d-1)$-cell code over any $|\Sigma| \geq 2$, fixing row 0 to a two-symbol beacon $(s_1, s_0, \ldots, s_0)$ $(s_1 \neq s_0)$ lets coordinate r read all d cells of row 0 (recovering $-c$ from the unique position v where $(\mathrm{rot}_c X)_{0,v} = s_1$) together with the anti-diagonal $\{(u, r-u) : u \in \mathbb{Z}_d\}$ (whose union with row 0 has size at most $2d-1$, containing $(c, r-c)$ where $(\mathrm{rot}_c X)_{c,r-c} = X_{c,r}$), leaving all $d(d-1)$ entries outside row 0 free. Part (b) combines Burnside/Möbius counting of aperiodic rot-orbits from [9, Appendix A] with divisor bounds showing that the non-free matrices occupy at most $d\,|\Sigma|^{d^2/2}$ states out of $|\Sigma|^{d^2}$, so $|C_d|/(|\Sigma|^{d^2}/d) \to 1$ along all $d \to \infty$ (prime or composite). □

Finally, at $M = 1$ and constant $\Delta \geq 3$ ([3, Problem 9.4(b)]), it remains open whether the pairing rate $\frac{1}{2} + o(1)$ holds or whether nonlinear bounded-arity shared codes beat $\frac{1}{2}$ between $\frac{1}{2}$ and $1 - p_{\min}(\Delta)$. In `BoundedAritySetCapacity.lean` we formalize the all-alphabet $\Delta = 2$ half-barrier question (`OneClassArityTwoHalfBarrierQuestion`, settled on paper for $\mathcal{E} = \emptyset$ in [3, Proposition 6.22(b)]) together with three canonical-alphabet (Σ_d) questions: the all-Δ half-barrier (`OneClassBoundedArityHalfBarrierQuestion`), bounded-arity coding gain above $\frac{1}{2}$ (`OneClassBoundedArityCodingGainAboveHalfQuestion`), and $1 - O(1/\Delta)$ inverse-arity scaling (`OneClassBoundedArityInverseArityScalingQuestion`).

Proposition 3.9 (Logical dichotomy for $M = 1, \Delta = O(1)$ capacity)**.** Lean: `oneClassBoundedArityCodingGainAboveHalf_iff_not_halfBarrier`, `oneClassCodingGain_of_inverseArityScaling` [`BoundedAritySetCapacityProofs.lean`]
Lean: `oneClassCanonicalArityTwo_of_allAlphabetHalfBarrier` [`SetCapacityConsequences.lean`]
Lean: `OneClassArityTwoHalfBarrierQuestion`, `OneClassBoundedArityHalfBarrierQuestion`, `OneClassBoundedArityCodingGainAboveHalfQuestion`, `OneClassBoundedArityInverseArityScalingQuestion` [`BoundedAritySetCapacity.lean`]

Over the canonical word alphabet $\Sigma_d = \{0,1\}^{\lceil \log_2 d \rceil}$, `OneClassBoundedArityCodingGainAboveHalfQuestion` *holds if and only if* `OneClassBoundedArityHalfBarrierQuestion` *fails,* `OneClassBoundedArityInverseArityScalingQuestion` *implies* `OneClassBoundedArityCodingGainAboveHalfQuestion`*, and the all-alphabet $\Delta = 2$ statement* `OneClassArityTwoHalfBarrierQuestion` *implies its Σ_d specialization (*`oneClassCanonicalArityTwo_of_allAlphabetHalfBarrier`*).*

Outline of the formal proof. The equivalence follows by quantifier negation between universal $\frac{1}{2}+\varepsilon$ upper bounds for all fixed $\Delta \geq 1$ and existence of $\Delta \geq 1, \varepsilon > 0$ exceeding rate $\frac{1}{2}+\varepsilon$ infinitely often; inverse-arity scaling $1 - C/\Delta$ exceeds $\frac{1}{2} + \frac{1}{4}$ once $\Delta > 4C$. □

4 Affine Hilbert cubes and Hales–Jewett-free composition

4.1 Progression limits vs. Hales–Jewett in two-stage composition

Suppose two linear class-uniform maps $K_1, K_2 : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ with class partitions $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ and $p_2 : \mathbb{Z}_d \to \mathrm{Fin}\, M_2$ satisfy $K_2 \circ K_1 = \tau$. By [9, Section 6, Lemma 6.1 and Theorem 6.3], any such factorization induces a common-message subspace $W \leq A$ inside a q-channel cyclic module $A \cong (\mathbb{Z}_d \to F^q)$ of dimension $\dim_F A \leq bM_2 d$ ($q \leq bM_2$) together with a linear readout $L : A \to Z$ injective on W such that $L(T_c w) = L(w)$ for all $w \in W$ and all c in a single p_1-class $J = p_1^{-1}(m_1)$ (where $(T_s w)_u = w_{u+s}$, formalized as `cyclicChannelShift`). When J contains a length-ℓ arithmetic progression $\{c_0 + js : 0 \leq j < \ell\}$, iterating T_s yields $\ell \dim_F W \leq \dim_F A + (\ell - 1)\dim_F \ker(T_s - \mathrm{id})$.

We record both the Hales–Jewett baseline and the Behrend/random progression barriers inside `BoundedArityDecoders` to make the quantitative bottleneck precise.

Proposition 4.1 (Hales–Jewett baseline vs. Behrend and random colorings)**.** Lean: `diagnosis_RamseyHalesJewettTwoStage`, `diagnosis_BehrendAndRandomAPBound` [`DiagnosisAndBarriersProofs.lean`]

Lean: `diagnosis5_CombinatorialInvVdWTendstoAndRankBound`, `diagnosis6_ModularBehrendAndRandomZModProgression` [`DiagnosisAndBridgesConsequences.lean`]

Lean: `Diagnosis_RamseyHalesJewettTwoStageSpec`, `Diagnosis_BehrendAndRandomAPBoundSpec` [`DiagnosisAndBarriers.lean`]

(a) Hales–Jewett progression baseline *([12]; [9, Section 6]): for every fixed $M_1 \geq 1$, the largest monochromatic progression length $\ell(d, M_1)$ guaranteed in every M_1-coloring of $\mathbb{Z}_d$ tends to ∞ as $d \to \infty$, ruling out exact two-stage linear factorizations at fixed M_1, M_2 for sufficiently large d.*

(b) Behrend and random progression limits *([2]): there exist 3-AP-free subsets of $\mathbb{Z}_d$ of size $d/2^{O(\sqrt{\log d})}$—hence M_1-colorings with $M_1 = 2^{O(\sqrt{\log d})}$ containing* no *non-constant 3-term progression modulo d—and 2-colorings of $\mathbb{Z}_d$ containing no monochromatic progression of length $\Omega(\log d)$.*

Outline of the formal proof. Part (a) derives divergence of the inverse van der Waerden threshold from Mathlib's Hales–Jewett theorem and plugs $\ell > M_2$ into the common-message dimension bound. Part (b) embeds base-B Euclidean spheres of $\{0, \ldots, \lfloor (B-1)/2 \rfloor\}^n$ into $\mathbb{Z}_d$ without wraparound carries ([2]) and bounds monochromatic progressions in random 2-colorings by a union bound. □

4.2 The Affine k-Cube Common-Message Growth Lemma

To dispense with arithmetic progressions over an *arbitrary* field F, consider any k linear endomorphisms $R_0, \ldots, R_{k-1} \in \mathrm{End}_F(A)$ (not even assumed to commute) and define for each $\varepsilon \in \{0,1\}^k$ the ordered operator cube vertex (`orderedCubeMap`)

$$R^{\varepsilon} \;:=\; R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0} \in \mathrm{End}_F(A).$$

Lemma 4.2 (Affine k-cube common-message growth)**.** Lean: `lemma_AffineCubeCommonMessageGrowth` [`AffineCubeCompositionProofs.lean`]
Lean: `Lemma_AffineCubeCommonMessageGrowthSpec` [`AffineCubeComposition.lean`]

Let F be any field, A a finite-dimensional F-vector space, Z an F-vector space, $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$, $W \le A$ a subspace, and $L : A \to Z$ an F-linear map injective on W. If $L(R^\varepsilon x) = L(x)$ for all $x \in W$ and all $\varepsilon \in \{0,1\}^k$, then

$$(k+1)\dim_F W \;\le\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

Outline of the formal proof. Define the nested cube-orbit subspaces $S_0 := W$ and $S_{t+1} := S_t + R_t(S_t)$ for $0 \le t < k$. By induction on t, every $u \in S_t$ is a linear combination of elements $R^\varepsilon x$ with $x \in W$ and $\varepsilon_s = 0$ for $s \ge t$. Appending $\varepsilon_t = 1$ versus $\varepsilon_t = 0$ in the cube invariance $L(R^\varepsilon x) = L(x)$ shows that $P_t := R_t - \mathrm{id}_A$ maps S_t into $S_{t+1} \cap \ker L$. Since $W \subseteq S_t \subseteq S_{t+1}$ and L is injective on W, we have $W \cap P_t(S_t) \subseteq W \cap \ker L = \{0\}$, so

$$\dim_F S_{t+1} \;\ge\; \dim_F W + \dim_F P_t(S_t) \;\ge\; \dim_F W + \dim_F S_t - \dim_F \ker(R_t - \mathrm{id}_A).$$

Summing from $t = 0$ to $k-1$ gives $(k+1)\dim_F W \le \dim_F S_k + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A) \le \dim_F A + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A)$. □

4.3 Elementary Cauchy–Schwarz cube extraction and Theorem 2A

Unlike arithmetic progressions, affine k-cubes are extracted in a single subset $S \subseteq \mathbb{Z}_d$ (even when S is 3-AP-free!) by elementary bucket pigeonholing and Cauchy–Schwarz, with explicit step bounds $1 \le s_t \le \lfloor d/(2k^2) \rfloor$ that guarantee $\dim_F \ker(T_{s_t} - \mathrm{id})$ stays small.

Lemma 4.3 (Cauchy–Schwarz affine k-cube extraction and dense-class bound)**.** Lean: `cauchySchwarzAffineCubeExtraction` [`AffineCubeCompositionProofs.lean`]
Lean: `singleSubset_CauchySchwarzAffineCubeExtraction`, `denseSubset_AffineCubeCommonMessageBound` [`AffineCubeConsequences.lean`]
Lean: `CauchySchwarzAffineCubeExtractionSpec` [`AffineCubeComposition.lean`]

Let $d, M_{\mathrm{eff}}, k \ge 1$ satisfy $\max\bigl(8M_{\mathrm{eff}}k^2,\ (2M_{\mathrm{eff}})^{2^k}\bigr) < d$.

(a) Single-subset form*: for every subset $J \subseteq \mathbb{Z}_d$ of density $|J| \ge d/M_{\mathrm{eff}}$ (regardless of whether J is 3-AP-free), there exist $c_0 \in J$ and steps $s_0, \dots, s_{k-1} \in \{1, \dots, \lfloor d/(2k^2) \rfloor\}$ such that $c_0 + \sum_{t:\varepsilon_t=1} s_t \in J$ for all $\varepsilon \in \{0,1\}^k$. Over any field F, any subspace $W \le (\mathbb{Z}_d \to F^q)$ with a linear readout injective on W and shift-invariant across J satisfies $(k+1)\dim_F W \le qd + qk\lfloor d/(2k^2) \rfloor$, ruling out exact two-stage linear factorizations whenever a single stage-1 class over an arbitrary label type κ_1 has size $\ge d/M_{\mathrm{eff}}$ at $k = M_2$ (*`denseSubset_AffineCubeCommonMessageBound`*).*

(b) Coloring form*: setting $M_{\mathrm{eff}} = M_1$ and $k = M_2$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ admits a monochromatic affine M_2-cube with steps in $\{1, \dots, \lfloor d/(2M_2^2) \rfloor\}$.*

Outline of the formal proof. Lift J to $S \subseteq \{0, \dots, d-1\}$. Partitioning $\{0, \dots, d-1\}$ into $Q = 2k^2$ intervals of length $\le \lfloor d/Q \rfloor + 1$ and applying Cauchy–Schwarz to the bucket sizes shows that some step $s \in \{1, \dots, \lfloor d/Q \rfloor\}$ satisfies $|S|(|S| - Q) \le 2d\,|\{x \in S : x + s \in S\}|$. Iterating k times shrinks the effective density denominator according to $D_{t+1} = 2D_t^2$ ($D_0 = M_{\mathrm{eff}}$, so $D_k = \frac{1}{2}(2M_{\mathrm{eff}})^{2^k}$); the threshold $d > \max(8M_{\mathrm{eff}}k^2, (2M_{\mathrm{eff}})^{2^k})$ keeps every iterate nonempty and prevents wraparound modulo d because $\sum_{t=0}^{k-1} s_t \le k\lfloor d/(2k^2) \rfloor < d$. Feeding the extracted cube into Lemma 4.2 with $\dim_F \ker(T_{s_t} - \mathrm{id}) \le q s_t \le q\lfloor d/(2k^2) \rfloor$ yields the common-message dimension bound. □

Theorem 4.4 (Theorem 2A: Hales–Jewett-free two-stage exact factorization cutoff)**.** Lean: `theorem2A_TwoStageRamseyFreeCutoff` [`AffineCubeCompositionProofs.lean`]
Lean: `theorem2A_GrowingClassesNonFactorization` [`AffineCubeConsequences.lean`]
Lean: `Theorem2A_TwoStageRamseyFreeCutoffSpec` [`AffineCubeComposition.lean`]

Let F be any *field, let $d, b, M_1, M_2 \geq 1$, and let $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. If two F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ satisfy $K_2 \circ K_1 = \tau$, then*

$$d \;\leq\; \max\Big(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\Big).$$

Consequently, for any growing class schedules $(M_1(d), M_2(d))$ eventually satisfying

$$\max\Big(8M_1(d)M_2(d)^2,\; (2M_1(d))^{2^{M_2(d)}}\Big) \;<\; d$$

—in particular for fixed M_2 and $M_1(d) = 2^{O(\sqrt{\log d})}$ on 3-AP-free Behrend partitions, or for fixed M_1 and $M_2(d) \leq (1-\varepsilon)\log_2\log_2 d$—no exact two-stage linear factorization $K_2 \circ K_1 = \tau$ exists over any field F.

Outline of the formal proof. Assume for contradiction that $\max(8M_1M_2^2, (2M_1)^{2^{M_2}}) < d$ and $K_2 \circ K_1 = \tau$. By Lemma 4.3(b), some p_1-class contains an affine M_2-cube with steps $1 \leq s_t \leq \lfloor d/(2M_2^2)\rfloor$. On the q-channel cyclic common-message module $A = (\mathbb{Z}_d \to F^q)$ ($q \leq bM_2$, $\dim_F A = qd \leq bM_2d$), the column-message reduction ([9, Lemma 6.1 and Theorem 6.3]) produces a subspace $W \leq A$ of dimension $\dim_F W = bd$ while $\dim_F \ker(T_{s_t} - \mathrm{id}) = q\gcd(s_t, d) \leq qs_t \leq bM_2\lfloor d/(2M_2^2)\rfloor$. Applying Lemma 4.2 with $k = M_2$ gives

$$(M_2+1)bd \;\leq\; bM_2d + \sum_{t=0}^{M_2-1} bM_2\lfloor d/(2M_2^2)\rfloor \;\leq\; bM_2d + \frac{1}{2}bd,$$

contradicting $bd > 0$. □

4.4 Sharpness at $d = 13$ and same-interval coded factorizations at $d = 4$

Two finite-dimensional controls establish that both halves of Lemma 4.2 and Theorem 4.4 are genuinely sharp.

Example 4.5 (Sharpness of $(k+1)$ vs. 2^k on an AP-free 2-cube at $d = 13$)**.** Lean: `control_D13TwoCubeAPFreeSharpness` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D13_GeneralFieldAndProgressionBridge` [`AffineCubeConsequences.lean`]
Lean: `Control_D13TwoCubeAPFreeSharpnessSpec` [`AffineCubeComposition.lean`]
Let $d = 13$, $q = 2$ ($\dim_F A = 26$), and consider the 2-cube $J_{13} = \{0,1,4,5\} = \{0,1\} + \{0,4\} \subseteq \mathbb{Z}_{13}$.

(a) J_{13} is 3-AP-free modulo 13 (`ThreeAPFreeMod`), so every non-constant progression in J_{13} has length $\ell \leq 2$, where the progression formula $\lfloor(26 + 2(\ell-1))/\ell\rfloor$ gives only $\dim_F W \leq 14$ (attained on $\{0,1\}$ over every field).

(b) Over *every* field F, Lemma 4.2 with $k = 2$ and $\dim_F \ker(T_s - \mathrm{id}) = 2$ ($s \in \{1,4\}$) gives $3\dim_F W \leq 26 + 2 + 2 = 30$, so $\dim_F W \leq 10 < 13$.

(c) Conversely, over $\mathbb{R}$—and more generally over every field F with $(13 : F) \neq 0$—there exists an explicit (W, L) invariant on J_{13} of dimension $\dim_F W = 9$. Since $4 \cdot 9 = 36 > 26 + 3 \cdot 2 = 32$, the naive 2^k-multiplier inequality $2^k \dim_F W \leq \dim_F A + (2^k - 1)q$ ($\dim_F W \leq 8$) is *false*.

Outline of the formal proof. Exhaustive residue check verifies J_{13} is 3-AP-free, and cyclic decomposition of $\mathbb{Z}_{13}$ gives $\dim_F \ker(T_s - \mathrm{id}) = 2$ for $s \not\equiv 0 \pmod{13}$ over every field. For the $D = 9$ witness, the constant frequency (2 dimensions) plus 7 non-constant Fourier modes whose 2×2 channel amplitudes satisfy a 3-constraint J_{13}-alignment system inside a 4-dimensional mode space contribute $2 + 7 = 9$ independent dimensions (realized over $\mathbb{R}$ via real/imaginary cosine-sine pairs and over general $(13 : F) \neq 0$ in `control_D13_GeneralFieldAndProgressionBridge`). □

Example 4.6 (Same-interval coded factorizations at $d = 4 = M_1 M_2$)**.** Lean: `control_D4SameIntervalCodedFactorization` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D4_AllWidthsAndBinaryFields` [`AffineCubeConsequences.lean`]
Lean: `Control_D4SameIntervalCodedFactorizationSpec` [`AffineCubeComposition.lean`]
Let $d = 4$ and let both stages use the *identical* two-interval partition $p_1 = p_2 = [0, 0, 1, 1]$ (`halfIntervalPartition4`, $M_1 = M_2 = 2$).

(a) *Uncoded impossibility*: for every lane width $b \geq 1$ and finite alphabet $|\Sigma| \geq 2$, neither entry-copy rules nor bit-lane-copy rules K_1, K_2 with $p_1 = p_2 = [0, 0, 1, 1]$ can satisfy $K_2 \circ K_1 = \tau$.

(b) *General-field coded factorization ($(20 : F) \neq 0$)*: over $\mathbb{R}$, and over every field F with $(20 : F) \neq 0$, for every lane width $b \geq 0$ there exist explicit F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{4\times 4}(F^b)$ with $p_1 = p_2 = [0, 0, 1, 1]$ satisfying exact transposition $K_2 \circ K_1 = \tau$.

(c) *Characteristic-2 and $(5 : F) \neq 0$ coded factorization across all widths (including $b = 1$ over $\mathbb{F}_2$)*: in characteristic 2 ($(2 : F) = 0$, including $\mathbb{F}_2$), a distinct pair of scalar 0/1 coefficient tables achieves $K_2 \circ K_1 = \tau$ lane-by-lane for *every* lane width $b \geq 0$ (including scalar $b = 1$ over $\mathbb{F}_2$). Combining $(2 : F) = 0$ and $(20 : F) \neq 0$ yields exact linear coded factorizations $K_2 \circ K_1 = \tau$ at all widths b over every field F with $(5 : F) \neq 0$ (`control_D4_AllWidthsAndBinaryFields`).

Outline of the formal proof. In (a), projecting onto a single lane reduces bit-lane copy to scalar entry copy over Σ, and pigeonholing the 4 entries of a row through 2-element interval classes forces a routing collision whenever linear coding across matrix positions is forbidden. In (b) and (c), linear combinations across entries within each 4×4 scalar lane disambiguate the two-element intervals $\{0, 1\}$ and $\{2, 3\}$: integer tables scaled by 20^{-1} satisfy the exact transpose identity whenever $(20 : F) \neq 0$, while separate binary 0/1 tables satisfy the exact transpose identity modulo 2 whenever $(2 : F) = 0$; any field with $(5 : F) \neq 0$ has either $(2 : F) = 0$ or $(20 : F) \neq 0$. □

5 Sumset polynomial recurrences and log-power rank deficits

5.1 From progressions to shifted monochromatic sumsets

Theorem 4.4 rules out *exact* factorizations $K_2 \circ K_1 = \tau$ up to exponentially growing M_1 (at fixed M_2), yet for *approximate* factorizations with constant class counts M_1, M_2—the regime relevant when trace cuts introduce $o(d^2)$ error—bounding the rank error

$$r \;:=\; \mathrm{rank}_F(\tau - K_2 \circ K_1)$$

via Hales–Jewett ([9, Theorem 6.6]) gave only an inverse-Ackermann/Hales–Jewett sub-iterated-logarithmic bound, because progressions in 2-colorings have length at most $O(\log d)$ and single-difference common-message arguments lose the entire remainder once $r \geq bd/\ell$.

Over any field F in which d is invertible ($(d : F) \neq 0$), we pass to a splitting field E/F carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions by $(2M_2)$-*fold monochromatic sumsets*

$$J_0 + B_1 + \cdots + B_{2M_2-1} \;\subseteq\; p_1^{-1}(m_1), \qquad |J_0| = k, \quad |B_t| = k+1.$$

For any finite $S \subseteq \mathbb{N}$, write $\mathrm{diam}_{\mathbb{N}}(S) := \max S - \min S$ (0 if $S = \emptyset$; `finsetNatDiam`). Shifting each B_t down by $b_{t,0} = \min B_t$ places $0 \in \widetilde{B}_t$ while preserving $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t)$, so *all* partial subset sums land inside $p_1^{-1}(m_1)$ simultaneously.

Lemma 5.1 (Minimum-shifting monochromatic sumsets and intrinsic diameter)**.** Lean: `unshiftedToShiftedSumset` [`SumsetRankDeficitProofs.lean`]
Lean: `finsetNatDiam_shift_min`, `unshiftedToShiftedSumset_intrinsicDiam` [`SumsetRankConsequences.lean`]
Lean: `UnshiftedToShiftedSumsetSpec` [`SumsetRankDeficit.lean`]

Given any unshifted monochromatic sumset $J_0 + B_1 + \cdots + B_n \subseteq p_1^{-1}(m_1)$ with $|J_0| = k$, $|B_t| = k+1$, and $B_t \subseteq [0, L_0]$, setting $\widetilde{B}_t := B_t - \min B_t$ and $\widetilde{J}_0 := J_0 + \sum_{t=1}^{n} \min B_t$ satisfies $0 \in \widetilde{B}_t$, $|\widetilde{B}_t| = k+1$, $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t) \leq L_0$, $|\widetilde{J}_0| = k$, and

$$\widetilde{J}_0 + \sum_{t \in S} \widetilde{B}_t \;\subseteq\; p_1^{-1}(m_1) \qquad \text{for every subset } S \subseteq \{1, \ldots, n\}.$$

Outline of the formal proof. Choosing $b_t \in \widetilde{B}_t$ for $t \in S$ and $0 \in \widetilde{B}_t$ for $t \notin S$ rewrites $c_0 + \sum_t \min B_t + \sum_{t \in S} b_t = c_0 + \sum_{t=1}^{n} (b_t + \min B_t)$ with $b_t + \min B_t \in B_t$. Translation by $-\min B_t$ on $\mathbb{N}_{\geq \min B_t}$ preserves cardinality and max − min. □

5.2 Arbitrary-subset Vandermonde pullback and sparse annihilating polynomials

Why did earlier Fourier/band arguments require J_0 to be a contiguous interval or arithmetic progression? Because standard Vandermonde matrices evaluate $1, z, \ldots, z^{k-1}$ at geometric progressions $\zeta^{c_0 + js}$! Over an arbitrary k-element base set $\widetilde{J}_0 \subseteq \mathbb{Z}_d$, the character-evaluation matrix $\big(\chi(\nu \cdot c)\big)_{c \in \widetilde{J}_0,\, 0 \leq \nu < k}$ is *already* a $k \times k$ Vandermonde matrix in the k distinct nodes $z_c = \chi(c)$ ($c \in \widetilde{J}_0$), because $\chi(\nu \cdot c) = \chi(c)^\nu$!

Lemma 5.2 (Arbitrary-subset Vandermonde injectivity and stacked band pullback)**.** Lean: `vandermondeArbitrarySubsetInjective`, `stackedBandPullbackInjective` [`SumsetRankDeficitProofs.lean`]
Lean: `VandermondeArbitrarySubsetInjectiveSpec`, `StackedBandPullbackInjectiveSpec` [`SumsetRankDeficit.lean`]

Let E be a field with an injective additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $J_0 \subseteq \mathbb{Z}_d$ satisfy $|J_0| = k \leq d$.

(a) For any E-vector space V, degree-$< k$ character evaluation $a \mapsto \big(\sum_{\nu=0}^{k-1} \chi(\nu c)\, a_\nu\big)_{c \in J_0}$ (`charBandEval`) is injective on V^k.

(b) For any two E-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d \times d}(E^b)$ with $J_0 \subseteq p_1^{-1}(m_1)$ and stage-1 class-m_1 decoder f_{1,m_1}, the frequency-$< k$ band space H (`SharedDecoders.bandRowSpace` at (χ, p_2, k), with $\dim_E H \leq bM_2k$) carries a pullback subspace $W \leq$ `SharedDecoders.ColumnAmbient`(H) satisfying

$$\dim_E W + \mathrm{rank}_E(\tau - K_2 \circ K_1) \;\geq\; bkd$$

such that the stacked evaluation $x \mapsto \big(f_{1,m_1}(\mathrm{rot}_c(\iota_H x))\big)_{c \in J_0}$ is injective on W.

Outline of the formal proof. In (a), $\chi(\nu c) = \chi(c)^\nu$, and injectivity of χ makes $\{\chi(c) : c \in J_0\}$ k distinct scalars in E; a degree-$< k$ vector-valued polynomial vanishing at k distinct points is identically zero. Part (b) restricts $\tau - K_2 \circ K_1$ to inputs whose columns lie in the k-frequency character subspace (bkd dimensions), intersects with $\ker(\tau - K_2 \circ K_1)$ to lose at most $r = \operatorname{rank}_E(\tau - K_2 \circ K_1)$ dimensions, and uses (a) to see that if $K_1 X$ vanishes on columns $c \in J_0$, then $(K_2 K_1 X)_{:,c} = (\tau X)_{:,c} = 0$ for all $c \in J_0$, forcing all k band coefficients of X to vanish. □

Next, each summand $\widetilde{B}_t$ of size $k+1$ supplies $k+1$ monomials against only k frequencies $\nu \in \{0, \dots, k-1\}$, guaranteeing a *nonzero* sparse polynomial supported on $\widetilde{B}_t$ that annihilates the entire frequency-$< k$ band while keeping small degree $\le \operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)$!

Lemma 5.3 (Sparse sumset annihilating polynomial)**.** Lean: `sumsetAnnihilatingPolynomial` [`SumsetRankDeficitProofs.lean`]
Lean: `sumsetAnnihilatingPolynomial_bandRowSpace` [`SumsetRankConsequences.lean`]
Lean: `SumsetAnnihilatingPolynomialSpec` [`SumsetRankDeficit.lean`]

Let $\chi : \mathbb{Z}_d \to E^\times$ be an injective additive character and let $\widetilde{B} \subseteq \mathbb{N}$ satisfy $0 \in \widetilde{B}$, $|\widetilde{B}| = k+1$, and $\max \widetilde{B} \le L_0$ (equivalently $\operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \le L_0$). Then there exists a nonzero polynomial $P \in E[X]$ supported in $\widetilde{B}$ with $\deg P \le \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \le L_0$ such that $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ for all $0 \le \nu < k$, and on any q-channel cyclic shift module ($\mathbb{Z}_d \to E^q$) (in particular on `SharedDecoders.ColumnAmbient(SharedDecoders.bandRowSpace(`χ, p_2, k`))`*, where $q \le bM_2k$),*

$$\dim_E \ker P(T_1) \;\le\; q \deg P \;\le\; bM_2kL_0.$$

Outline of the formal proof. The homogeneous linear system $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ ($0 \le \nu < k$) has k equations in $|\widetilde{B}| = k+1$ unknowns, hence a nonzero solution P supported in $\widetilde{B}$ of degree $\le \max \widetilde{B} = \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \le L_0$. On a q-channel cyclic sequence $w : \mathbb{Z}_d \to E^q$, $P(T_1)w = 0$ fixes w uniquely from any $\deg P$ consecutive coordinates once $\deg P \le d$ (and trivially $\dim \le qd \le q \deg P$ if $\deg P > d$), giving $\dim_E \ker P(T_1) \le q \deg P$. □

5.3 Two-stage sumset rank bound, Kővári–Sós–Turán extraction, and Theorem 2B

Applying $2M_2 - 1$ annihilating polynomials $P_t(T_1)$ sequentially to the pullback W grows an ambient chain of $2M_2$ subspaces inside a bM_2kd-dimensional ambient space, forcing the error rank r to scale linearly in bkd:

Theorem 5.4 (Two-stage sumset recurrence rank bound)**.** Lean: `sumsetTwoHopRankBound` [`SumsetRankDeficitProofs.lean`]
Lean: `SumsetTwoHopRankBoundSpec` [`SumsetRankDeficit.lean`]

Let F be a field with $(d : F) \neq 0$, let $M_1, M_2, b \ge 1$, and let $1 \le k \le d$. Suppose $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a shifted $(2M_2)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ in class m_1 with $\sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \le d$. Then for every $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and every pair of F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$, writing $r := \operatorname{rank}_F(\tau - K_2 \circ K_1)$,

$$2r \;\ge\; bk\Big(d - \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t\Big), \qquad \text{and if } \mathrm{diamB}_t \le \lfloor d/(4M_2) \rfloor \text{ for all } t, \quad 4r \;\ge\; bkd.$$

Outline of the formal proof. Since $(d : F) \neq 0$, $X^d - 1$ is separable over F; passing to its splitting field E/F preserves $\operatorname{rank}(\tau - K_2 \circ K_1)$ and supplies an injective character $\chi : \mathbb{Z}_d \to E^\times$. Let $W \le A_{\mathrm{band}}$

($\dim_E A_{\mathrm{band}} \le bM_2kd$) be the pullback from Lemma 5.2(b) on $\widetilde{J}_0$, with $\dim_E W \ge bkd - r$. For each $0 \le t < 2M_2 - 1$, let P_t be the sparse annihilating polynomial on $\widetilde{B}_t$ from Lemma 5.3. Because $\chi(\nu \cdot -)$ is annihilated by P_t while every shift in $\widetilde{J}_0 + \sum_{s \le t} \widetilde{B}_s$ stays in $p_1^{-1}(m_1)$ where K_1 agrees with $\tau \pm$ error on the error-free subspace, $P_t(T_1)$ maps the t-th span $U_t := \sum_{S \subseteq \{0,\dots,t-1\}} (\prod_{s \in S} T_{\widetilde{B}_s}) W$ into the kernel of stacked $\widetilde{J}_0$-evaluation, which intersects W trivially! Hence each of the $2M_2 - 1$ steps adds at least $\dim_E W - bM_2k \,\mathrm{diamB}_t$ dimensions inside A_{band}:

$$2M_2 \dim_E W - bM_2k \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \;\le\; \dim_E A_{\mathrm{band}} \;\le\; bM_2kd.$$

Substituting $\dim_E W \ge bkd - r$ gives $2M_2(bkd - r) \le bM_2k(d + \sum_t \mathrm{diamB}_t)$, i.e., $2r \ge bk(d - \sum_t \mathrm{diamB}_t)$. When every $\mathrm{diamB}_t \le \lfloor d/(4M_2) \rfloor$, $\sum_t \mathrm{diamB}_t \le (2M_2 - 1)\frac{d}{4M_2} \le \frac{d}{2}$, giving $4r \ge bkd$. □

Lemma 5.5 (Iterated Kővári–Sós–Turán monochromatic sumset extraction)**.** Lean: `iteratedKovariSosTuranSumset` [`SumsetRankDeficitProofs.lean`]
Lean: `IteratedKovariSosTuranSumsetSpec` [`SumsetRankDeficit.lean`]

For every $M_1, M_2 \ge 1$, there exist $0 < c_{\mathrm{lo}} \le C_{\mathrm{hi}}$ and $D_0 \in \mathbb{N}$ such that for all $d \ge D_0$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ contains an unshifted (and hence shifted) $(2M_2)$-fold monochromatic sumset in $[0, \lfloor d/(4M_2) \rfloor]$ at width scale $1 \le k \le d$ satisfying

$$c_{\mathrm{lo}} (\log d)^{\frac{1}{2M_2-1}} \;\le\; k \;\le\; C_{\mathrm{hi}} (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Choose the densest color class m_1 inside $[0, \lfloor d/2 \rfloor)$ and build a bipartite addition table between blocks of length $\le \lfloor d/(4M_2) \rfloor$ and base intervals. At each of the $2M_2 - 1$ stages, Kővári–Sós–Turán [14] (formalized via falling-factorial double counting of $(k+1)$-tuples of common neighbors) extracts k_{t+1} left vertices and $k + 1$ right shifts at exponential-in-k^{2M_2-1} density cost, balancing at $k = \Theta_{M_1,M_2}((\log d)^{1/(2M_2-1)})$. □

Theorem 5.6 (Theorem 2B: log-power two-stage rank deficit over admissible fields)**.** Lean: `theorem2B_TwoStageLogPowerRankDeficit` [`SumsetRankDeficitProofs.lean`]
Lean: `theorem2B_ExplicitConstant`, `theorem2BExplicitConst_pos` [`SumsetRankConsequences.lean`]
Lean: `Theorem2B_TwoStageLogPowerRankDeficitSpec` [`SumsetRankDeficit.lean`]

For all integers $M_1, M_2 \ge 1$, define the explicit positive constant and threshold (`theorem2BExplicitConst`*,* `theorem2BExplicitThreshold`*)*

$$c(M_1, M_2) \;:=\; \frac{1}{144\,(\log(16M_1M_2))^{\frac{1}{2M_2-1}}} \;>\; 0,$$

$$D_0(M_1, M_2) \;:=\; \max\Big(4M_2,\; \lceil \exp(36^{2M_2-1} \log(16M_1M_2)) \rceil + 2\Big).$$

Then for every field F, every $d \ge D_0(M_1, M_2)$ with $(d : F) \ne 0$, every lane width $b \ge 1$, and every pair of F-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 classes,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\ge\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Combine Lemma 5.5 (evaluated with explicit KST parameters at $k = \lfloor (\frac{\log d}{18^{2M_2-1} \log(16M_1M_2)})^{1/(2M_2-1)} \rfloor$) and Theorem 5.4 ($r \ge bkd/4$). □

Corollary 5.7 (Single-class second stage: $\Omega(b\,d\log d)$ rank deficit)**.** Lean: `theorem2B_OneSecondStage_OmegaLog` [`SumsetRankConsequences.lean`]

When $M_2 = 1$ ($2M_2 - 1 = 1$), for every $M_1 \geq 1$ there exist $c > 0$ and $D_0 \geq 2$ such that for every field F, every $d \geq D_0$ with $(d : F) \neq 0$, every $b \geq 1$, and every M_1-class K_1 and 1-class K_2 on $\mathrm{Mat}_{d\times d}(F^b)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c\,b\,d\log d.$$

Outline of the formal proof. Specialize Theorem 5.6 to $M_2 = 1$, where $\frac{1}{2M_2-1} = 1$. □

5.4 Single-class band relaxation obstruction at $d = 12$ and the macroscopic target

Why did Theorem 5.4 use the *actual* stage-1 class-uniform equality $(K_2K_1X)_{:,c} = (\tau X)_{:,c}$ across shifted sumsets rather than merely relaxing stage 1 to an abstract linear readout L on `SharedDecoders.ColumnAmbient`(H) for a single class $J \subseteq \mathbb{Z}_d$ of density $1/2$ and a single $k = 2$ band? Because that abstract single-class band relaxation is *false*, even over $\mathbb{C}$ at $d = 12$!

Example 5.8 (Unconditional $d = 12$ single-class band witness over $\mathbb{C}$)**.** Lean: `control_D12SingleClassAbstractBand` [`SumsetRankDeficitProofs.lean`]

Lean: `exists_primitive_twelfth_root_complex`, `d12AddChar_injective_and_primitive`, `control_D12_UnconditionalExists`, `control_D12_BandRowSpaceChannelEquiv` [`SumsetRankConsequences.lean`]

Lean: `Control_D12SingleClassAbstractBandSpec` [`SumsetRankDeficit.lean`]

At $d = 12$ and $k = 2$ with $q = 4$ channels (realized e.g. at $bM_2 = 2$ so $q = bM_2k = 4$), on the half-density subset $J_{12} = \{0, 1, 3, 4, 7, 9\} \subseteq \mathbb{Z}_{12}$, there exist unconditionally a primitive 12-th root of unity $\zeta_{12} = \exp(2\pi i/12) \in \mathbb{C}$, a shift-intertwining linear equivalence between $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ and `SharedDecoders.ColumnAmbient`$($`SharedDecoders.bandRowSpace`$(\chi, p_0, 2))$ at lane width $b = 2$ and constant 1-class partition $p_0(c) = 0$ ($M_2 = 1$), a linear readout $L : (\mathbb{Z}_{12} \to \mathbb{C}^4) \to \mathbb{C}^{12}$, and a full 24-dimensional subspace W ($\dim_{\mathbb{C}} W = 24 = kd$) on which stacked evaluation over J_{12} is injective and realizes *every* pair of band vectors $(a_0, a_1) \in (\mathbb{C}^{12})^2$ as $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for all $c \in J_{12}$.

Outline of the formal proof. Define $\zeta_{12} = \exp(2\pi i/12) \in \mathbb{C}$, verify $\zeta_{12}^{12} = 1$ and $\zeta_{12}^m \neq 1$ for $1 \leq m < 12$, and build the explicit 24-dimensional 4-channel encoding over $\mathbb{Z}_{12}$ whose J_{12}-shifts decode $a_0 + \zeta_{12}^c a_1$ exactly; `control_D12_BandRowSpaceChannelEquiv` identifies $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ shift-equivariantly with `SharedDecoders.ColumnAmbient` of `SharedDecoders.bandRowSpace` at $b = 2, M_2 = 1, k = 2$. □

Consequently, pushing two-stage linear rank deficits from $\Omega(bd(\log d)^{1/(2M_2-1)})$ all the way to $\varepsilon(M_1, M_2)bd^2$ must couple *multiple* stage-1 classes or *multiple* frequency bands simultaneously. We formalize both the general-field macroscopic target (`OpenTarget_MacroscopicTwoStageRankDeficitSpec`) and its $(d : F) \neq 0$ specialization (`OpenTarget_MacroscopicTwoStageRankDeficitAdmissibleFieldSpec`).

Proposition 5.9 (Macroscopic two-stage rank target and low-error refutation)**.** Lean: `openTarget_admissible_of_unrestricted` [`SumsetRankDeficitProofs.lean`]

Lean: `conditional_NoLowErrorTwoStageLinear_of_MacroscopicTarget` [`MachineSweepConsequences.lean`]

Lean: `OpenTarget_MacroscopicTwoStageRankDeficitSpec`, `OpenTarget_MacroscopicTwoStageRankDeficitAdmissibleFieldSpec` [`SumsetRankDeficit.lean`]

`OpenTarget_MacroscopicTwoStageRankDeficitSpec` *implies* `OpenTarget_MacroscopicTwoStageRankDeficitAdmissibleFieldSpec`, *and under* `OpenTarget_MacroscopicTwoStageRankDefici`

`tSpec`*, for every fixed* $M_1, M_2 \geq 1$ *there exists* $\varepsilon > 0$ *such that along every sequence of fields* F_d*, lane widths* $b_d \geq 1$*, and* M_1, M_2*-class linear stages* $K_{1,d}, K_{2,d}$*, the normalized rank error* $\operatorname{rank}_{F_d}(\tau - K_{2,d} \circ K_{1,d})/(b_d d^2)$ *cannot converge to* 0 *as* $d \to \infty$*.*

Outline of the formal proof. Immediate from the quantifier specialization and contrapositive of the rank lower bound. □

Part II
Operator entanglement, swap codes, and native sweep extraction

6 Anti-diagonal untwisting and operator entanglement duality

6.1 Pairwise shift collisions, anti-diagonal tensor separation, and mask floors

Problem 9.1 of [3] (together with the characteristic-split constructions and blow-ups of [7]) asks what structural constraints govern shared decoders on aperiodic class partitions $p : \mathbb{Z}_d \to \kappa$, while [9, Theorem 3.4 and Appendix C] proved the universal capacity bound $\dim_F C \leq \frac{41}{64} bd^2 + \frac{5}{3} bd^{3/2}$ $(M = 2)$ alongside probabilistic-method floors showing that random-mask certificate arguments stall at 9/16 (full-ambient isolation) and 5/8 (nested component counting).

Our first observation is that *every* shift-collision condition on a linear source $C \leq \mathrm{Mat}_{d\times d}(A)$ decouples into d completely independent 1-dimensional slices once we untwist matrix coordinates along anti-diagonals $a = c + j \in \mathbb{Z}_d$. Specifically, define the mutually inverse F-linear maps (`antiDiagLinear`, `fromAntiDiagLinear`)

$$Z_{c,a} \;:=\; X_{c,\,a-c}, \qquad X_{c,j} \;=\; Z_{c,\,j+c} \qquad (c, a, j \in \mathbb{Z}_d),$$

and the explicit tensor isomorphism $\Psi = \texttt{antiDiagTensorEquiv} : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$ sending $u \otimes v$ to $(c, j) \mapsto u_c\, v_{j+c}$.

Theorem 6.1 (Pairwise shift-collision characterization and anti-diagonal tensor separation)**.** Lean: `pairwiseShiftCollisionCharacterization`, `antiDiagonalTensorSeparation` [`OperatorDualityProofs.lean`]

Lean: `antiDiagTensorEquiv_intertwines_rot`, `shiftSetCovariance_tensor_finrank` [`DualityAndSwapConsequences.lean`]

Lean: `PairwiseShiftCollisionCharacterizationSpec`, `AntiDiagonalTensorSeparationSpec` [`OperatorDuality.lean`]

Let F *be a field,* $d \geq 1$*,* A *an* F*-vector space, and* $p : \mathbb{Z}_d \to \kappa$*.*

(a) Pairwise shift-collision characterization*: a linear subspace* $C \leq \mathrm{Mat}_{d\times d}(A)$ *admits deterministic (possibly nonlinear)* p*-shared decoders (*`SharedDecoders.IsSharedCode`*) if and only if for every shift* $s \in \mathbb{Z}_d$ *and every* $Y \in C$ *with* $\mathrm{rot}_s(Y) \in C$*,* Y *satisfies pairwise shift covariance (*`InPairwiseShiftCovariance`*): whenever* $p(c) = p(c + s)$*,* $Y_{c+s,\,j} = Y_{c,\,j+s}$ *for all* $j \in \mathbb{Z}_d$*.*

(b) Right-factor untwisting of rot_t*: under* $\Psi : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$*, column rotation* rot_t *intertwines with the pure right-factor coordinate shift* $\mathrm{id} \otimes \sigma^t$*, where* $(\sigma^t v)_a = v_{a+t}$*.*

(c) Left-factor untwisting of shift-link covariance*: for any shift set $E \subseteq \mathbb{Z}_d$, let $\sim_E$ denote the equivalence relation generated by* `SwapRigidity.LinkpE` *and let $U_E(p) \le (\mathbb{Z}_d \to F)$ be the subspace of vectors constant on $\sim_E$-components (*`leftComponentSubmodule`*). Then Ψ maps $U_E(p) \otimes_F (\mathbb{Z}_d \to A)$ onto the shift-set covariance subspace $H_E(p)$ (*`shiftSetCovarianceSubmodule`*), and when $\dim_F A < \infty$,*

$$\dim_F H_E(p) \;=\; \dim_F U_E(p) \cdot d \cdot \dim_F A.$$

Outline of the formal proof. In (a), determinism of $f_{p(c)}$ on C is equivalent by linearity of C to saying that whenever $X_1, X_2 \in C$ satisfy $p(c_1) = p(c_2)$ and $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$, rows c_1 of X_1 and c_2 of X_2 agree; setting $s = c_2 - c_1$ and $Y = X_1 - \mathrm{rot}_{-s}(X_2) \in C$ gives $\mathrm{rot}_s(Y) = \mathrm{rot}_s(X_1) - X_2 \in C$ and the stated covariance. In (b), $\Psi(\mathrm{id} \otimes \sigma^t(u \otimes v))_{c,j} = u_c v_{j+c+t} = \mathrm{rot}_t(\Psi(u \otimes v))_{c,j}$. In (c), writing $X_{c,j} = Z_{c,j+c}$ turns $X_{c',j+\delta'} = X_{c,j+\delta}$ under $c + \delta = c' + \delta'$ into $Z_{c',a} = Z_{c,a}$ for every anti-diagonal $a = c + \delta + j$, independently across $a \in \mathbb{Z}_d$. □

Proposition 6.2 (Mask-certificate floors at 9/16 and 5/8 and generic tensor realization)**.** Lean: `diagnosis_LinearSharedDecoderBoundsAndMaskFloors` [`DiagnosisAndBarriersProofs.lean`]
Lean: `diagnosis4_ExplicitSquareCoverAndMaskCertificateBridge` [`DiagnosisAndBridgesConsequences.lean`]
Lean: `maskFloor_GenericSubspaceRealization` [`DualityAndSwapConsequences.lean`]
Lean: `Diagnosis_LinearSharedDecoderBoundsAndMaskFloorsSpec` [`DiagnosisAndBarriers.lean`]
Let F be a field, $d \ge 1$, and A a finite-dimensional F-vector space of dimension $b := \dim_F A \ge 1$.

(a) Universal capacity vs. random-mask certificate floors *([9, Theorems 3.4, 3.5, C.6, and C.7]): every 2-class linear shared code satisfies $\dim_F C \le \frac{41}{64} bd^2 + \frac{5}{3} bd^{3/2}$, and every M-class linear shared code ($1 \le M, 2M \le d$) has rate $\le 1 - \frac{1}{4\lceil\sqrt{2M}\rceil}$ (*`ceilSqrtTwoMul`*); meanwhile, on uniformly random binary masks $p : \mathbb{Z}_d \to \mathrm{Bool}$ ($\eta \ge 0$), the full-ambient isolation certificate exceeds $(\frac{9}{16} - \frac{\eta}{4} + \frac{1}{2d})bd^2$ outside probability $(d-1)e^{-2\eta^2 d/9}$ and the nested-component certificate exceeds $(\frac{5}{8} - \frac{\eta}{4})bd^2$ outside probability $(d-1)e^{-\eta^2 d/2}$.*

(b) Single-shift generic tensor realization of the 5/8 floor*: for any shift $s \in \mathbb{Z}_d$, any equal-dimension pair $C_0, C_s \le \mathrm{Mat}_{d\times d}(A)$ with $C_0 \cap C_s \le H_{\{0,s\}}(p)$ obeys $2\dim_F C_0 \le d^2 \dim_F A + \dim_F H_{\{0,s\}}(p)$, and generic anti-diagonal tensor subspaces $C_0 = \Psi(V_0 \otimes \top)$, $C_s = \Psi(V_s \otimes \top)$ attain $C_0 \cap C_s = H_{\{0,s\}}(p)$ with $2\dim_F C_0 + d\dim_F A \ge d^2 \dim_F A + \dim_F H_{\{0,s\}}(p)$—reaching rate $\ge \frac{5}{8} - \frac{\eta}{4} - \frac{1}{2d}$ whenever* `SharedDecoders.componentCount`$(p, \{0, s\}) \ge (\frac{1}{4} - \frac{\eta}{2})d$.

(c) Alternating-mask separation across shifts*: on $d = 2m$ ($m \ge 2$) with alternating $p(c) = (c \bmod 2 == 1)$, shift $s = 1$ has $p(c) \ne p(c+1)$ everywhere so $U_{\{0,1\}}(p) = \top$ and $H_{\{0,1\}}(p) = \top$ (allowing single-shift rate 1), whereas shift $s = 2$ has $\dim_F U_{\{0,2\}}(p) = 2$ and forces every true shared code C to satisfy $2\dim_F C \le (d^2 + 2d)\dim_F A$ (rate $\le \frac{1}{2} + \frac{1}{d}$).*

Outline of the formal proof. Part (a) bridges [9]'s universal capacity theorems and Hoeffding random-mask concentration bounds with explicit integer square-cover arithmetic on `ceilSqrtTwoMul`. Part (b) splits $(\mathbb{Z}_d \to F) = U_{\{0,s\}}(p) \oplus W_{\mathrm{comp}}$ and assigns complementary $\lfloor \dim W_{\mathrm{comp}}/2 \rfloor$-dimensional halves to V_0 and V_s before tensoring with $(\mathbb{Z}_d \to A)$. Part (c) checks that odd shifts never link equal parities while shift 2 links all even indices into one component and all odd indices into a second component. □

6.2 Operator Schmidt Rank separates transpose from Stone's perfect shuffle

We now turn to [3, Problem 9.5]: why can a multi-tape Turing machine execute Stone's perfect shuffle [15] in $O(L)$ steps, whereas matrix transposition is conjectured to require $\Omega(L \log L)$ steps, even though both permute bit addresses of $\mathrm{Mat}_{d\times d}(F^b)$?

View $\texttt{GridState}(F, d, b) = (\mathrm{Fin}\, d \to \mathrm{Fin}\, d \to \mathrm{Fin}\, b \to F)$ as the tensor product $F^d_{\mathrm{row}} \otimes_F (F^d_{\mathrm{col}} \otimes_F F^b)$ separating the *coarse row register* $r \in \mathrm{Fin}\, d$ from the *within-row column-and-lane register* $(c, \ell) \in \mathrm{Fin}\, d \times \mathrm{Fin}\, b$. An F-linear operator K on $\texttt{GridState}(F, d, b)$ has *Operator Schmidt Rank* $\le R$, written $\mathrm{rk}_\otimes(K) \le R$ (`HasOperatorSchmidtRankLe`), if there exist R row matrices $A_i \in \mathrm{Mat}_{d\times d}(F)$ and R within-row maps $B_i \in \mathrm{End}_F(F^d \otimes F^b)$ such that

$$(KX)_{r,c,\ell} \;=\; \sum_{i=0}^{R-1} \sum_{r'=0}^{d-1} (A_i)_{r,r'}\, \big(B_i(X_{r'})\big)_{c,\ell},$$

and *exact* Operator Schmidt Rank $\mathrm{rk}_\otimes(K) = R$ (`HasExactOperatorSchmidtRank`) if R is minimal.

Lemma 6.3 (Submultiplicativity and rank-1 locality)**.** Lean: `operatorSchmidtRankSubmultiplicative` [`OperatorDualityProofs.lean`]
Lean: `isRowOrColUniform_rankLe_one` [`DualityAndSwapConsequences.lean`]
Lean: `OperatorSchmidtRankSubmultiplicativeSpec` [`OperatorDuality.lean`]

Over any field F:

(a) If $\mathrm{rk}_\otimes(K_1) \le R_1$ and $\mathrm{rk}_\otimes(K_2) \le R_2$, then $\mathrm{rk}_\otimes(K_2 \circ K_1) \le R_1 R_2$.

(b) Every row-uniform column operator $I_d \otimes B$ (`IsRowUniformColOp`) and every column-uniform row operator $A \otimes I_{db}$ (`IsColUniformRowOp`) satisfies $\mathrm{rk}_\otimes(K) \le 1$.

Outline of the formal proof. Expanding $(K_2 \circ K_1)$ across pairs $(i_2, i_1) \in \mathrm{Fin}\, R_2 \times \mathrm{Fin}\, R_1$ multiplies the row matrices $A_{2,i_2} A_{1,i_1}$ and composes the column operators $B_{2,i_2} \circ B_{1,i_1}$, using at most $R_1 R_2$ elementary Kronecker terms; row-uniform and column-uniform maps are single elementary tensors $I_d \otimes B$ and $A \otimes \mathrm{id}$. □

Theorem 6.4 (Exact Operator Schmidt Rank: $\mathrm{rk}_\otimes(\tau) = d^2$ vs. $\mathrm{rk}_\otimes(S) = 4$)**.** Lean: `operatorSchmidtRankSwap`, `operatorSchmidtRankStoneShuffle` [`OperatorDualityProofs.lean`]
Lean: `OperatorSchmidtRankSwapSpec`, `OperatorSchmidtRankStoneShuffleSpec` [`OperatorDuality.lean`]

Let F be any field and $b \ge 1$.

(a) Matrix transpose *(`gridSwapLinear`): for every $d \ge 1$, $\mathrm{rk}_\otimes(\tau) = d^2$.*

(b) Stone's perfect shuffle and unshuffle *(`stoneShuffleGridLinear`, `stoneUnshuffleGridLinear`): on $d = 2^m$, S and S^{-1} are mutual inverses with $\mathrm{rk}_\otimes(S) \le 4$ and $\mathrm{rk}_\otimes(S^{-1}) \le 4$ for all $m \ge 0$, and for every $m \ge 1$ ($d = 2^m \ge 2$),*

$$\mathrm{rk}_\otimes(S) \;=\; \mathrm{rk}_\otimes(S^{-1}) \;=\; 4.$$

Outline of the formal proof. For (a), the d^2 matrix units $E_{c,r} \otimes E^{\mathrm{lane}\,0}_{r,c}$ decompose τ into d^2 terms, and evaluating any R-term decomposition on row-localized basis inputs exhibits a spanning set of size R for the d^2-dimensional space of $d \times d$ matrices (via linear independence of the d^2 partial-transpose slices), forcing $R \ge d^2$.

For (b), when $d = 2^m$ ($m \geq 1$), Stone's 1-bit left cyclic address rotation sends cell (r, c) to (r', c') where the high bit of r moves into the low bit of c' and the high bit of c moves into the low bit of r'. Splitting on the $2 \times 2 = 4$ parities/halves (low bit of c', low bit of r') $\in \{0,1\}^2$ writes $S = \sum_{u,v\in\{0,1\}} A_{u,v} \otimes B_{u,v}$ in 4 Kronecker terms, and linear independence of those 4 non-zero quadrant-halving/doubling blocks when $2^m \geq 2$ proves no $R' < 4$ suffices (and symmetrically for S^{-1}). □

Theorem 6.5 (Pass lower bounds for Stone-shuffle and bounded-OSR pipelines)**.** Lean: `stoneShufflePipelinePassLowerBound` [`OperatorDualityProofs.lean`]
Lean: `boundedOSRPipelinePassLowerBound` [`DualityAndSwapConsequences.lean`]
Lean: `StoneShufflePipelinePassLowerBoundSpec` [`OperatorDuality.lean`]

Let F be any field and $b \geq 1$.

(a) On $d = 2^m$ ($m \geq 0$), any linear operator pipeline (`composeGridOps`) composed of arbitrarily many row-uniform column operators (`IsRowUniformColOp`), arbitrarily many column-uniform row operators (`IsColUniformRowOp`), and t Stone shuffle or unshuffle passes that equals τ satisfies

$$(2^m)^2 \;\leq\; 4^t \qquad \textit{and hence} \qquad t \;\geq\; m = \log_2 d.$$

(b) More generally, for any $d \geq 1$ and $R \geq 2$, any pipeline realizing τ with t operators of $\mathrm{rk}_\otimes \leq R$ and arbitrarily many operators of $\mathrm{rk}_\otimes \leq 1$ satisfies $d^2 \leq R^t$ and $t \geq \dfrac{2 \log d}{\log R}$.

Outline of the formal proof. By Lemma 6.3 and Theorem 6.4(b), the composed pipeline has $\mathrm{rk}_\otimes \leq 4^t$ (respectively $\mathrm{rk}_\otimes \leq R^t$), whereas $\mathrm{rk}_\otimes(\tau) = d^2$ by Theorem 6.4(a). Monotonicity of log yields $t \geq \log_2 d$ and $t \geq 2 \log d / \log R$. □

To complement Theorem 6.5 on actual machines, we construct native $O(L)$-time `Transpose.TM` implementations of Stone's shuffle, Stone's unshuffle, and within-row shear, confirming that the $O(1)$-vs.-$\log_2 d$ separation is operationally real on multi-tape Turing machines:

Proposition 6.6 (Native $O(L)$-time Turing machines for shuffle, unshuffle, and row shear)**.** Lean: `diagnosis_StoneShuffleAndRowShearMachineComplexity`, `stoneUnshuffleTM_twoTape_linear` [`DiagnosisAndBarriersProofs.lean`]
Lean: `Diagnosis_StoneShuffleAndRowShearMachineComplexitySpec` [`DiagnosisAndBarriers.lean`]

(a) Stone's shuffle on $d = 2^m$ ($m \geq 1$) is computed in $O(L)$ steps on a 3-work-tape `Transpose.TM`, Stone's unshuffle on $d = 2^m$ ($m \geq 1$) is computed in $O(L)$ steps on a 2-work-tape `Transpose.TM` (`stoneUnshuffleTM_twoTape_linear`), and within-row cyclic shear $X_{r,c} \mapsto X_{r,(c+r) \bmod d}$ ($d \geq 1$) is computed in $O(L)$ steps on a 2-work-tape `Transpose.TM`.

(b) Over every field F, for all $m \geq 1$ ($d = 2^m$), all $b \geq 1$, and every dyadic column-bit scale $j \in \mathrm{Fin}\, m$, the commutator $[S, I_{\mathrm{row}} \otimes G_j]$ of a single Stone shuffle pass S (`stoneShuffleGridLinear`) with the j-th column-bit flip involution G_j (`colBitFlipGridLinear`) has exact macroscopic rank $\mathrm{rank}_F(SG_j - G_jS) = b(2^m)^2/2$ simultaneously across all $m = \log_2 d$ scales (cf. [3, Section 4.3, Corollary 6.13, and Remark 6.14]).

Outline of the formal proof. Because d and $b = \lceil \log_2 d \rceil$ are not hardcoded into the fixed machine, the shuffle machine first discovers unary counters for b and $H = d^2/2$ across tapes 1 and 2 in $O(L)$ steps, buffers the first H input words on tape 0, and interleaves b-bit words from tape 0 and the

second input half. The 2-work-tape unshuffle machine keeps the unary b-counter on tape 1, emits even-indexed b-bit words directly from the input tape to the output tape while buffering odd-indexed words on tape 0, and then replays tape 0 to output. Row shear discovers d and b and cyclically rotates each db-bit row by rb bits. For (b), $SG_j - G_jS$ pairs off complementary bit-j/high-bit cell orbits of size 2 on half the grid entries, giving exact rank $b(2^m)^2/2$. □

6.3 Adversarial controls: row-diagonal gauges and skew one-time pads

Two sharp controls prevent over-interpreting $\mathrm{rk}_\otimes$ outside its domain of validity:

(i) A 2-tape machine can execute within-row shear $(K_{\mathrm{shear}}X)_{r,c} = X_{r,c+r}$ in $O(L)$ steps (Proposition 6.6(a)) because rows never talk to each other, yet K_{shear} applies a *different* operator on each row r, giving $\mathrm{rk}_\otimes(K_{\mathrm{shear}}) = d$! We show that replacing $\mathrm{rk}_\otimes \le 1$ by the full row-diagonal subalgebra $\mathcal{D}_{\mathrm{row}} \otimes \mathrm{End}_F(F^d \otimes F^b)$ (`IsRowDiagonalOp`) preserves the $s \ge \log_2 d$ pass lower bound via *output-row source arity* (`HasOutputRowSourceArityLe`).

(ii) For *nonlinear* intermediate encodings on work tapes, cell-wise Shannon mutual information with target columns can vanish identically even when $W \oplus X = X^\top$!

Proposition 6.7 (Row-shear $\mathrm{rk}_\otimes = d$ vs. row-diagonal gauge invariance)**.** Lean: `control_RowShearAndRowDiagonalAlternation` [`OperatorDualityProofs.lean`] Lean: `rowDiagonal_SourceArityGaugeInvariance` [`DualityAndSwapConsequences.lean`] Lean: `Control_RowShearAndRowDiagonalAlternationSpec` [`OperatorDuality.lean`]

Let F be any field, $d \ge 1$, and $b \ge 1$.

(a) Within-row shear (`rowShearLinear`) belongs to `IsRowDiagonalOp` and has exact Operator Schmidt Rank $\mathrm{rk}_\otimes(K_{\mathrm{shear}}) = d$.

(b) Output-row source arity (`HasOutputRowSourceArityLe`) is gauge-invariant under two-sided composition $D_1 \circ K \circ D_0$ with arbitrary row-diagonal operators D_0, D_1.

(c) On $d = 2^m$, every alternating product $D_sU_sD_{s-1}\cdots U_1D_0$ (`alternatingRowDiagProduct`) of $s+1$ arbitrary row-diagonal operators D_i and s Stone shuffle/unshuffle passes $U_i \in \{S, S^{-1}\}$ has output-row source arity $\le 2^s$; if it equals τ, then $2^m \le 2^s$ and $s \ge m = \log_2 d$.

Outline of the formal proof. In (a), $K_{\mathrm{shear}} = \sum_{r=0}^{d-1} E_{r,r} \otimes \sigma^r$, and linear independence of the d distinct column shifts $\sigma^0, \ldots, \sigma^{d-1}$ forces $\mathrm{rk}_\otimes = d$. In (b) and (c), row-diagonal operators transform each row in place (multiplying source-row sets by 1), whereas each S or S^{-1} reads at most 2 source rows per output row, bounding source arity by 2^s; since row 0 of $\tau(X)$ depends on all 2^m rows of X, $2^m \le 2^s$. □

Proposition 6.8 (Skew one-time-pad zero-mutual-information trap)**.** Lean: `control_SkewOneTimePadInformationTrap` [`OperatorDualityProofs.lean`] Lean: `uniformShannonMI_eq_zero_of_indep`, `skewOneTimePad_ShannonMutualInfoZero` [`DualityAndSwapConsequences.lean`] Lean: `Control_SkewOneTimePadInformationTrapSpec` [`OperatorDuality.lean`]

For $d, b \ge 1$, define $W(X) := X^\top \oplus X$ on $\Sigma_b^{d\times d}$ ($\Sigma_b = \{0,1\}^b$, `skewXorPad`). Then $W(X) \oplus X = X^\top$ for all X, every diagonal cell $W_{r,r}$ is identically 0, and under uniformly random $X \in \Sigma_b^{d\times d}$, every single cell $W_{r,c}$ is statistically independent of every target column $X_{:,c'}$—so its discrete Shannon mutual information (`uniformShannonMI`) satisfies

$$I(W_{r,c}\,;\, X_{:,c'}) \;=\; 0 \qquad \text{for all } r, c, c' \in \mathbb{Z}_d.$$

Outline of the formal proof. When $r = c$, $W_{r,r} = 0$ is deterministic. When $r \neq c$, at least one of the two distinct matrix entries $X_{c,r}$ and $X_{r,c}$ lies outside column c' (since (c, r) and (r, c) have distinct column indices $r \neq c$!), acting as an independent uniform one-time pad on the other and making the joint law of $(W_{r,c}, X_{:,c'})$ a product distribution of zero mutual information. □

7 Three-term swap codes and multi-scale dyadic masks

7.1 Disproving the $d(d+1)/2$ pairing bound for three-term swap codes (Remark 5.6)

In [3, Theorem 5.1 and Corollary 5.5], any subspace $S \leq \mathrm{Mat}_{d\times d}(F)$ on which a *two*-Kronecker-term operator $K(X) = A_1XB_1 + A_2XB_2$ computes the transpose ($K(X) = X^\top$ for all $X \in S$) has dimension at most the symmetric pairing bound $d(d+1)/2$, via complete invariant flags and triangularization of a matrix pencil over the algebraic closure. Immediately afterward, [3, Remark 5.6] observed that at $d = 3$ and $d = 4$ a three-term operator exceeds $d(d+1)/2$, and remarked: *"We have no three-term example for $d \geq 5$."*

We resolve this question over *every* field F (arbitrary characteristic, with no root-of-unity or characteristic hypotheses) by constructing an explicit block-Kronecker family K_{3m} on $d = 3m$ (`threeTermBlockA`, `threeTermBlockB`) together with diagonal tail padding (`threeTermPaddedA`, `threeTermPaddedB`). Write the three unordered off-diagonal pairs of $\{0, 1, 2\}$ as $(i_0, j_0) = (0, 1)$, $(i_1, j_1) = (0, 2)$, and $(i_2, j_2) = (1, 2)$, and set

$$A_t \;:=\; (E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m, \qquad B_t \;:=\; -A_t \;=\; (E_{j_t,i_t} - E_{i_t,j_t}) \otimes I_m \qquad (t \in \{0, 1, 2\}).$$

Theorem 7.1 (Exact dimension $4m(m+1)$ of the three-term Kronecker family K_{3m})**.** Lean: `bonus1_ThreeTermBlockSwapCodeDim` [`SwapCodeAnswersProofs.lean`]
Lean: `Bonus1_ThreeTermBlockSwapCodeDimSpec` [`SwapCodeAnswers.lean`]
Let F be any *field and let $m \geq 1$. Partition $X \in \mathrm{Mat}_{3m\times 3m}(F)$ into a 3×3 grid of $m \times m$ blocks $(X_{i,j})_{0\leq i,j<3}$.*

(a) X lies in `SwapRigidity.swapCode`$(A, B) = \{X : \sum_{t=0}^{2} A_tXB_t = X^\top\}$ *if and only if* every *block $X_{i,j}$ is symmetric ($X_{i,j}^\top = X_{i,j}$ for all $0 \leq i, j < 3$) and the three diagonal blocks sum to zero ($X_{0,0} + X_{1,1} + X_{2,2} = 0$;* `IsThreeByThreeSymmetricZeroTraceBlocks`*).*

(b) Consequently, over every field F,

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;=\; 8 \cdot \frac{m(m+1)}{2} \;=\; 4m(m+1).$$

Since $4m(m+1) - \frac{3m(3m+1)}{2} = \frac{m(5-m)}{2}$, this strictly exceeds the pairing bound $d(d+1)/2$ ($d = 3m$) for $m \in \{1, 2, 3, 4\}$ ($d = 3 : 8 > 6$; $d = 6 : 24 > 21$; $d = 9 : 48 > 45$; $d = 12 : 80 > 78$).

Outline of the formal proof. Expanding $\sum_{t=0}^{2}(E_{i_t,j_t} - E_{j_t,i_t})X(E_{j_t,i_t} - E_{i_t,j_t})$ blockwise sends each off-diagonal block (i, j) $(i \neq j)$ to $X_{j,i}$ (coming from the unique pair $\{i, j\}$) and sends each diagonal block (i, i) to $\sum_{j\neq i}(-X_{j,j})$. Setting this equal to the block transpose $(X_{j,i}^\top)_{i,j}$ requires $X_{j,i} = X_{j,i}^\top$ for all $i \neq j$, plus $-X_{1,1} - X_{2,2} = X_{0,0}^\top$, $-X_{0,0} - X_{2,2} = X_{1,1}^\top$, and $-X_{0,0} - X_{1,1} = X_{2,2}^\top$, which is equivalent over *every* ring (including characteristic 2!) to $X_{i,i}^\top = X_{i,i}$ for $i \in \{0, 1, 2\}$ together with $X_{0,0} + X_{1,1} + X_{2,2} = 0$ (since subtracting any two diagonal equations gives $X_{0,0}^\top - X_{1,1}^\top = X_{1,1} - X_{0,0}$,

etc., and substituting $X_{2,2} = -X_{0,0} - X_{1,1}$ closes the system). Eight independent symmetric $m \times m$ blocks (6 off-diagonal plus 2 free diagonal blocks) each have dimension $m(m+1)/2$, giving $8 \cdot \frac{m(m+1)}{2} = 4m(m+1)$. □

Embedding the $3m \times 3m$ construction in the top-left corner of $\mathrm{Mat}_{d\times d}(F)$ and reusing term $t = 2$ (A_2, B_2) to place $+1$ on the remaining $d - 3m$ diagonal entries yields odd and non-multiple-of-3 counterexamples, including $d = 7$ $(29 > 28)$:

Theorem 7.2 (Padded three-term swap codes and unified disproof across $d \in \{3, 4, 6, 7, 9, 12\}$)**.**
Lean: `bonus1_ThreeTermPaddedSevenAndFive` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus1_UnifiedThreeTermExceedsPairingBound` [`DualityAndSwapConsequences.lean`]
Lean: `Bonus1_ThreeTermPaddedSevenAndFiveSpec`, `OpenQuestion_ThreeTermDimFiveExceedsPairingSpec` [`SwapCodeAnswers.lean`]

Over every *field F:*

(a) Padding $m = 2$ $(3m = 6)$ to $d = 7$ achieves exact dimension $29 > 28 = \frac{7\cdot 8}{2}$; padding $m = 1$ $(3m = 3)$ to $d = 4$ achieves $11 > 10 = \frac{4\cdot 5}{2}$; and padding $m = 1$ to $d = 5$ achieves $15 = \frac{5\cdot 6}{2}$.

(b) Consequently, for every field F and every $d \in \{3, 4, 6, 7, 9, 12\}$, there exist three-term operators $A, B : \mathrm{Fin}\, 3 \to \mathrm{Mat}_{d\times d}(F)$ with

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;>\; \frac{d(d+1)}{2}.$$

Whether $d = 5$ admits any field F and three-term operator exceeding $5 \cdot 6/2 = 15$ remains open and is formalized as `OpenQuestion_ThreeTermDimFiveExceedsPairingSpec`*.*

Outline of the formal proof. Because term $t = 2$ carries $(E_{1,2} - E_{2,1}) \otimes I_m$ on the top-left $3m \times 3m$ block and I_{d-3m} on the tail block, cross-terms between the active $(1, 2)$ blocks and the tail vanish unless the corresponding $m \times (d - 3m)$ blocks satisfy coupled transpose equations, contributing $24 + 5 = 29$ dimensions at $(m, d) = (2, 7)$, $8 + 3 = 11$ at $(1, 4)$, and $8 + 7 = 15$ at $(1, 5)$. Combining $(1, 4)$ and $(2, 7)$ with Theorem 7.1 covers all $d \in \{3, 4, 6, 7, 9, 12\}$. □

7.2 Multi-scale dyadic intervals vs. periodic Walsh masks (Problem 9.3)

Problem 9.3 of [3] asks what swap-code dimension can be achieved on $d = 2^m$ when *both* left multipliers A_a and right multipliers B_a lie in spans of rotations masked across *all* dyadic scales simultaneously. Inspecting the geometry of scale-j masks reveals a fundamental split between two natural meanings of "scale-j dyadic masks":

(i) *Contiguous dyadic / cyclic interval masks* (`IsAlignedDyadicInterval`, `IsCyclicInterval`): each mask I of length 2^{m-1-j} has at most 2 **cyclic boundaries** on $\mathbb{Z}_{2^m}$, so $M_A(j)$ masks at scale j introduce at most $2\sum_j M_A(j)$ total endpoints!

(ii) *Periodic Rademacher/Walsh digit masks* (`walshDigitMaskRotMatrix`, `IsPeriodicScaleDyadicMask`): a single bit mask $\{r : \mathrm{bit}_j(r) = 1\}$ has 2^{j+1} **cyclic boundaries**, so the $m = \log_2 d$ Walsh bit masks already refine $\mathrm{Fin}(2^m)$ into 2^m singleton atoms!

We first resolve the contiguous dyadic/cyclic interval formulation completely over every field F.

Theorem 7.3 (Multi-scale interval swap bound, full-profile attainment, and $\frac{1}{2} + o(1)$ limit)**.** Lean: `bonus2_MultiscaleIntervalBothSides`, `bonus2_MultiscaleFullProfileAttainsFullSwap` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus2_SublinearProfileAndTendstoHalf` [`DualityAndSwapConsequences.lean`]
Lean: `Bonus2_MultiscaleIntervalBothSidesSpec`, `Bonus2_MultiscaleFullProfileAttainsFullSwapSpec` [`SwapCodeAnswers.lean`]
Let F be any field, $m \geq 0$, and $d = 2^m$.

(a) Every aligned dyadic interval-rotation span (`InAlignedDyadicIntervalRotSpan`*) is a multi-scale cyclic-interval span (*`InMultiscaleIntervalRotSpan`*), and whenever finite families $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ lie in* `InMultiscaleIntervalRotSpan` *with mask-count profiles $M_A, M_B : \mathrm{Fin}\, m \to \mathbb{N}$, setting $\beta_A := \max(1,\, 2\sum_j M_A(j))$ and $\beta_B := \max(1,\, 2\sum_j M_B(j))$,*

$$2\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; d^2 + d\min(\beta_A, \beta_B).$$

(b) Sublinear profiles*: whenever $\min(\sum_j M_{A,m}(j),\, \sum_j M_{B,m}(j)) = o(2^m)$ as $m \to \infty$, the upper bound $\frac{1}{2} + \frac{\min(\beta_A,\beta_B)}{2d}$ converges to $\frac{1}{2}$, so $\dim_F(\texttt{SwapRigidity.swapCode}\ A\ B) \leq (\frac{1}{2} + o(1))d^2$.*

(c) Full dyadic profile attainment*: for the complete aligned profile $M_{\mathrm{full}}(j) = 2^{j+1}$ ($\sum_{j=0}^{m-1} 2^{j+1} \geq 2^m$ for $m \geq 1$), aligned dyadic interval-rotation families attain full swap dimension d^2:*

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;=\; d^2.$$

Outline of the formal proof. In (a), all cyclic intervals occurring in A introduce at most $2\sum_j M_A(j)$ boundary cuts on the cycle $\mathbb{Z}_{2^m}$, partitioning $\mathrm{Fin}(2^m)$ into at most β_A contiguous atomic intervals on each of which every A_a acts uniformly as a linear combination of rotations! Thus $\sum_a A_a X B_a = X^\top$ is β_A-class-uniform on rows (giving $2\dim \leq d^2 + d\beta_A$ by the translate-pigeonhole bound [3, Theorem 6.4]) and, by transposing the equation to $\sum_a B_a^\top X A_a^\top = X^\top$, symmetrically β_B-class-uniform on columns (giving $2\dim \leq d^2 + d\beta_B$). Part (b) follows by dividing by $2d^2$. In (c), indicator differences of aligned dyadic intervals down to length $2^0 = 1$ span all singleton point projectors $E_{r,r}$, realizing all matrix units $E_{r,c} = E_{r,r}\sigma^{c-r}$ and the full swap. □

We then formalize the exact structural invariants explaining why the periodic Rademacher/Walsh formulation of [3, Problem 9.3] defeats interval boundary counting yet spans only a tiny $(m+1)2^m$-dimensional operator algebra per side:

Theorem 7.4 (Periodic Rademacher/Walsh boundary and operator-span dichotomy)**.** Lean: `walshDigitMask_in_periodicDyadicSpan`, `openTarget_walsh_of_periodicProfile` [`SwapCodeAnswersProofs.lean`]
Lean: `walshBitSet_cyclicBoundaryCount`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`, `bonus2_WalshDichotomyStructuralProperties` [`DualityAndSwapConsequences.lean`]
Lean: `OpenTarget_MultiscalePeriodicWalshProfileSpec`, `OpenTarget_MultiscalePeriodicWalshSwapCodeSpec` [`SwapCodeAnswers.lean`]
On $d = 2^m$ ($m \geq 1$):

(a) Every cyclic interval $I \subseteq \mathrm{Fin}(2^m)$ has cyclic boundary count `cyclicBoundaryCount`$(I) \leq 2$*, whereas the scale-j Walsh bit set $W_j = \{r : bit_j(r) = 1\}$ (*`walshBitSet`*) has exact cyclic boundary count* `cyclicBoundaryCount`$(W_j) = 2^{j+1}$ *(and every* `IsPeriodicScaleDyadicMask` *at scale j has cyclic boundary count 2^{j+1}).*

(b) *The m Walsh bit sets $W_0, \dots, W_{m-1}$ jointly separate all 2^m points of* $\mathrm{Fin}(2^m)$ *into 2^m singleton atoms, making one-sided class-refinement bounds vacuous.*

(c) *Nevertheless, over every field F, the entire Walsh-masked rotation algebra spanned by $\{P_{W_j,b}\sigma^s : j \in \mathrm{Fin}\, m,\ b \in \{0,1\},\ s \in \mathrm{Fin}(2^m)\}$ lies inside* `InMultiscalePeriodicDyadicRotSpan` *with profile $M(j) = 1$ ($\sum_j M(j) = m = \log_2 d \ll d$) and has operator dimension at most $(m+1)2^m \ll 4^m$.*

(d) *Consequently, the periodic-profile bound* `OpenTarget_MultiscalePeriodicWalshProfileSpec` *implies the Walsh digit-span bound* `OpenTarget_MultiscalePeriodicWalshSwapCodeSpec`.

Outline of the formal proof. In (a), W_j alternates runs of 2^{m-1-j} zeros and 2^{m-1-j} ones 2^j times around $\mathrm{Fin}(2^m)$, crossing $2 \cdot 2^j = 2^{j+1}$ boundaries, whereas any single interval crosses at most 2 endpoints. Part (b) reads off the m-bit binary expansion of $r \in \mathrm{Fin}(2^m)$. In (c), $P_{W_j,0} + P_{W_j,1} = I_{2^m}$, so the $2m$ masks $\{P_{W_j,0}, P_{W_j,1}\}$ are spanned by I_{2^m} together with the m masks $\{P_{W_j,1}\}_{j<m}$; tensoring with the 2^m cyclic shifts σ^s spans at most $(m+1)2^m$ dimensions. Part (d) instantiates $M_A(j) = M_B(j) = 1$ once $m \le \delta 2^m$. □

8 Native periodic sweep extraction and the frontier toward (⊤)

8.1 Cut-cone determination and symbolic transition-table trees

Problem 9.4 of [3] asks how a real Turing machine's finite transition table $M.\delta$ turns bounded-depth trace cuts into shared bounded-arity decoders on nonlinear input fibres. In `NativeSweepInterface.lean` and `MachineSweepConsequences.lean`, we formalize both the abstract cut-cone determination principle and two concrete symbolic tree extraction theorems directly from the step function of `Transpose.TM`.

Let M be a native `Transpose.TM` with $1 \le M.k$ (available without loss of generality for any transposer via Lemma 2.2), and let $C \subseteq \Sigma_d^{d\times d}$ be a frozen trace/cut fibre (`IsFrozenTraceCutFibre`) over horizon T with head trajectories (inPos, workPos), emission schedule emitStep, and cut step set $\Gamma \subseteq \{0, \dots, T-1\}$.

Lemma 8.1 (Cut-causal cone determination on frozen fibres)**.** Lean: `tmTraceCutConeDetermination` [`NativeSweepInterfaceProofs.lean`]
Lean: `evalConeTree_traceConeTree_eq` [`MachineSweepConsequences.lean`]
Lean: `TMTraceCutConeDeterminationSpec` [`NativeSweepInterface.lean`]

(a) *On* `IsFrozenTraceCutFibre`*, for every step $v < T$, the control state and scanned work-tape symbols of M at step v on $X \in C$ are uniquely determined by the input bits read at steps w in the Γ-stopped causal cone of v (*`InCutCausalCone`*).*

(b) *When every Γ-free causal path in $\{0, \dots, T-1\}$ has length $\le D$ ($1 \le D$), evaluating the symbolic depth-D* `CausalConeTree` *extracted relative to column c (*`traceConeTree`*) on $\mathrm{rot}_c(X)$ via $M.\delta$ (*`evalConeTree`*) reproduces the exact machine state, scanned work symbols, and emitted output bit $X_{c,r,\ell}$ for every $X \in C$, while reading at most $(M.k+1)^D$ input cells per bit and at most $\Delta = b(M.k+1)^D$ input cells per b-bit output word.*

Outline of the formal proof. Induct on the depth budget $m \le D$. At any step $w < T$, the control state comes either from step 0 (initial state 0), from $w - 1 \in \Gamma$ (frozen across C), or recursively

from $w-1 \notin \Gamma$; similarly, scanned work-tape cell $h \in \mathrm{Fin}(M.k)$ comes either from blank initial tape (0), from its most recent visit $u < w$ if $u \in \Gamma$ (frozen across C), or recursively from $u \notin \Gamma$. Since any path not hitting Γ terminates within D steps, depth D reaches a leaf on every branch, and each internal node branches $(M.k+1)$-ways while performing one relative input-bit read on $\mathrm{rot}_c(X)$. □

8.2 Two native periodic sweep extraction theorems

When head sweeps over columns c in the same schedule class $p_0(c) \in \mathrm{Fin}\, M_0$ execute isomorphic relative causal trees outside an irregular schedule set $\mathcal{E}_{\mathrm{sched}}$, cut steps $u \in \Gamma$ touched by an output cone can be handled in two complementary ways: either (A) absorb cut-touching output cells into $\mathcal{E}$ while keeping exact class count M_0, or (B) keep *zero* extra broken cells ($\mathcal{E} = \mathcal{E}_{\mathrm{sched}}$) by recording at most L_{cut} frozen cut-slot labels per column in the refined class index!

Theorem 8.2 (Cut-absorbed native periodic sweep extraction)**.** Lean: `tmPeriodicSweepExtraction_cutAbsorbed` [`MachineSweepConsequences.lean`]
Lean: `periodicSweepInducesBoundedAritySharedCode`, `periodicSweepNonlinearFibreDeficit` [`NativeSweepInterfaceProofs.lean`]
Lean: `PeriodicSweepInducesBoundedAritySharedCodeSpec`, `PeriodicSweepNonlinearFibreDeficitSpec` [`NativeSweepInterface.lean`]

Let M be a `Transpose.TM` *with $1 \le M.k$, let $d \ge 1$ and $D \ge 1$, and let $C \subseteq \Sigma_d^{d\times d}$ satisfy* `IsFrozenTraceCutFibre` *with Γ-free causal path length $\le D$. Suppose outside $\mathcal{E} := \mathcal{E}_{\mathrm{sched}} \cup$ `cutTouchingOutputCells`$(d, T, \mathrm{workPos}, \mathrm{emitStep}, \Gamma)$, output cones in class $p_0(c) \in \mathrm{Fin}\, M_0$ share cut-free relative tree templates* $\mathrm{template}(p_0(c), r, \ell)$*. Then with $\Delta := b(M.k+1)^D$, evaluating* template *via $M.\delta$ induces an M_0-class* `IsBoundedAritySetCode` *on C with broken set $\mathcal{E}$, and whenever $1 \le M_0$ and $C \ne \emptyset$,*

$$\frac{\log|C|}{\log|\Sigma_d|} \;\le\; d^2 - (d^2 - |\mathcal{E}| - dM_0\Delta)\frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}}.$$

Outline of the formal proof. For $(r,c) \notin \mathcal{E}$, no bit cone of (r,c) touches Γ, so `traceConeTree` coincides with `cutFreeTraceConeTree`, which equals $\mathrm{template}(p_0(c), r, \ell)$ by periodicity. Lemma 8.1(b) verifies the `IsBoundedAritySetCode` recovery identity and support bound $\Delta = b(M.k+1)^D$, and Theorem 3.4 gives the capacity bound. □

Theorem 8.3 (Cut-label-refined native periodic sweep extraction)**.** Lean: `tmPeriodicSweepExtraction_cutLabelRefined` [`MachineSweepConsequences.lean`]

Under the hypotheses of Theorem 8.2, suppose instead that for $(r,c) \notin \mathcal{E}_{\mathrm{sched}}$, every cut step $u \in \Gamma$ touched by a bit cone of (r,c) lies among L_{cut} column-indexed cut slots $\mathrm{colCutStep}(c) : \mathrm{Fin}\, L_{\mathrm{cut}} \to \mathbb{N}$*, and the symbolic slot trees* `traceSlotConeTree` *equal* $\mathrm{slotTemplate}(p_0(c), r, \ell)$*. Refining p_0 by column c's tuple of L_{cut} frozen cut labels in $\mathrm{Fin}(M.q+1) \times \mathrm{Fin}(M.g+1)^{M.k}$ yields*

$$M_{\mathrm{cls}} \;:=\; M_0 \cdot \big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$$

classes and an explicit $(\Delta, p_{\mathrm{ref}}, \mathcal{E}_{\mathrm{sched}})$-bounded-arity set code on C ($\Delta = b(M.k+1)^D$, with zero *extra broken cells beyond $\mathcal{E}_{\mathrm{sched}}$) satisfying*

$$\frac{\log|C|}{\log|\Sigma_d|} \;\le\; d^2 - (d^2 - |\mathcal{E}_{\mathrm{sched}}| - dM_{\mathrm{cls}}\Delta)\frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}}$$

whenever $1 \le M_0$ and $C \ne \emptyset$.

Outline of the formal proof. Substituting the fibre's constant cut labels $\mathrm{cutVal}(\mathrm{colCutStep}(c,s))$ into the leaves of $\mathrm{slotTemplate}(p_0(c), r, \ell)$ reconstructs `traceConeTree` for every $(r,c) \notin \mathcal{E}_{\mathrm{sched}}$. Encoding the pair $(p_0(c),\ s \mapsto \mathrm{cutVal}(\mathrm{colCutStep}(c,s)))$ into $\mathrm{Fin}\, M_{\mathrm{cls}}$ produces p_{ref} and invokes Theorem 3.4. □

8.3 Conditional closures and the open frontier toward (T)

Question (T)—formalized in its negated linear-time form as `TapeCodingConjectureNegationSpec` ($\neg$ `Transpose.LinearTimeTransposable`, paired with `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ for $d \geq 2$)—remains open. What our theorems change is that the artificial nonlinear-set, Hales–Jewett, and Stone-shuffle walls identified in [3, 9] are gone: Theorem 3.4 already forces a strictly positive rate deficit $\Omega(1/\Delta)$ on nonlinear frozen-cut fibres whenever a trace cut yields sublinearly many periodic cone classes $M_{\mathrm{cls}} = o(d/\Delta)$ and advice budget smaller than that deficit! We formalize two clean conditional closures of (T) capturing the two main routes forward:

Theorem 8.4 (Conditional closures of (T) from periodic sweep extraction or the $M = 1$ half-barrier)**.** Lean: `conditional_TapeCodingConjecture_of_PeriodicCellSweepHypothesis`, `conditional_TapeCodingConjecture_of_HalfBarrierAndOneClassFibre` [`MachineSweepConsequences.lean`]

Lean: `TapeCodingConjectureNegationSpec` [`NativeSweepInterface.lean`]

(a) Direct $\Theta(1/\Delta)$-deficit route*: under* `PeriodicCellSweepReductionHypothesis`*—asserting that every $1 \leq k$-work-tape $c(L+1)$-step transposer admits a constant cone arity $\Delta \geq 1$, an unbroken output fraction $\delta \in (0,1]$, a sublinear class count $M_{\mathrm{cls}}(d) = o(d/\Delta)$, and a trace/cut advice budget $B_{\mathrm{adv}}(d) \leq \varepsilon c(L+1)$ with $\varepsilon < \frac{\delta}{8c}\, p_{\min}(\Delta)$ whose pigeonhole fibre $C_d \subseteq \Sigma_d^{d\times d}$ is a $(\Delta, p_d, \mathcal{E}_d)$-bounded-arity set code for infinitely many d—the negation of linear-time transposability (*`TapeCodingConjectureNegationSpec`*) holds unconditionally (including for 0-work-tape machines via Lemma 2.2).*

(b) $M = 1$ half-barrier route*: if 1-class constant-arity set codes obey the $\frac{1}{2} + o(1)$ pairing rate (*`OneClassBoundedArityHalfBarrierQuestion`*) and every $1 \leq k$-work-tape $O(L)$ transposer admits 1-class ($\mathcal{E} = \emptyset$) constant-arity fibres of rate $\geq 3/4$ infinitely often (*`OneClassSweepFibreReductionHypothesis`*), then* `TapeCodingConjectureNegationSpec` *holds.*

Outline of the formal proof. In (a), dummy-tape padding (Lemma 2.2) upgrades any M_0 with $c_0(L+1)$ runtime to $1 \leq M.k$ and $1 \leq c$. Choosing d large enough that $M_{\mathrm{cls}}(d)\Delta/d < \delta/2$, Theorem 3.7(a) caps $\log_2|C_d| \leq (1 - \frac{\delta}{2} p_{\min}(\Delta))L$, whereas pigeonholing $2^L \leq 2^{B_{\mathrm{adv}}(d)}|C_d|$ gives $\log_2|C_d| \geq (1 - 2c\varepsilon)L$, contradicting $4c\varepsilon < \frac{\delta}{2} p_{\min}(\Delta)$. In (b), instantiate $\varepsilon = 1/8$ in `OneClassBoundedArityHalfBarrierQuestion` to contradict rate $\geq 3/4 > 5/8$. □

We close by recording the four sharp open problems to which this paper reduces the gap around (T):

Open Target 1. The $M = 1, \Delta \geq 3$ arbitrary-set half-barrier (`OneClassBoundedArityHalfBarrierQuestion`, Proposition 3.9; cf. [3, Problem 9.4(b)]): for windows $\Delta \geq 3$ on spread relations over Σ_d, does a single shared deterministic rule obey rate $\frac{1}{2} + o(1)$, or does nonlinear bounded-arity coding beat $\frac{1}{2}$ below $1 - p_{\min}(\Delta)$ (e.g. below $\frac{229}{256}$ at $\Delta = 3$)? (At $\Delta \leq 2$ with $\mathcal{E} = \emptyset$, a written proof of $\frac{1}{2} + \frac{3}{2d}$ over all alphabets was given in [3, Proposition 6.22(b)], leaving the Lean verification of `OneClassArityTwoHalfBarrierQuestion` and extension to $\mathcal{E} \neq \emptyset, M = o(d)$.)

Open Target 2. **Macroscopic $\varepsilon(M_1, M_2)bd^2$ two-stage linear rank deficit** (`OpenTarget_MacroscopicTwoStageRankDeficitSpec`, Proposition 5.9): can multi-class / multi-band coupling overcome the single-class $d = 12$ obstruction (Example 5.8) and upgrade Theorem 5.6's $\Omega(bd(\log d)^{1/(2M_2-1)})$ error rank to $\Omega_{M_1,M_2}(bd^2)$ (including over $\mathbb{F}_2$ at $d = 2^m$)?

Open Target 3. **All-scale periodic Rademacher/Walsh swap codes** (`OpenTarget_MultiscalePeriodicWalshSwapCodeSpec`, Theorem 7.4; cf. [3, Problem 9.3]): on $d = 2^m$, when both sides lie in the $(m+1)2^m$-dimensional span of Walsh digit masks and rotations, is $\dim_F \texttt{SwapRigidity.swapCode}(A, B) \le c_{\text{walsh}}\, 4^m$ for a uniform constant $c_{\text{walsh}} < 1$?

Open Target 4. **Sub-deficit advice compression for adaptive/periodic trace cuts** (`PeriodicCellSweepReductionHypothesis`, Theorem 8.4(a); cf. [3, Problem 9.4]): in Theorems 8.2 and 8.3, arbitrary adaptive head trajectories on an $O(L)$-step machine have $2^{\Theta(L)}$ naive step-by-step schedules, exceeding the $\Theta(L/\Delta)$ bit deficit of Theorem 3.4 when $\Delta = (k+1)^D$ is large. Compressing coarse sweep schedules and cut-slot templates below $\frac{\delta}{8c}\, p_{\min}(\Delta)\, L$ bits would trigger Theorem 8.4(a) directly.

A Formalization architecture and boundary

The `BoundedArityDecoders` Lean 4 package imports the repository's existing libraries without modifying any upstream library, namely `Transpose` ([5]), `BoundedStackDepth` ([4]), `TapeStorageRates` ([10]), `SwapRigidity` ([3]), `AdaptiveTransfer` ([11]), and `SharedDecoders` ([9]), all pinned to Lean and Mathlib version `4.34.1` [16].

The formalization is organized into twenty modules across three layers:

(1) *Seven formal specification modules* (`BoundedAritySetCapacity.lean`, `AffineCubeComposition.lean`, `SumsetRankDeficit.lean`, `OperatorDuality.lean`, `SwapCodeAnswers.lean`, `NativeSweepInterface.lean`, and `DiagnosisAndBarriers.lean`), defining 51 named ': Prop' specifications covering every theorem, lemma, finite control, prior-spine benchmark, and open target of the paper;

(2) *Seven primary proof modules* (`*Proofs.lean`), establishing complete kernel-checked proofs of every proved specification; and

(3) *Six consequence and representation-bridge modules* (`SetCapacityConsequences.lean`, `AffineCubeConsequences.lean`, `SumsetRankConsequences.lean`, `DualityAndSwapConsequences.lean`, `MachineSweepConsequences.lean`, and `DiagnosisAndBridgesConsequences.lean`), proving the unconditional empty/nonempty set extensions, explicit constants, all-dimension asymptotics, Mathlib `TensorProduct` intertwining equivalences, discrete Shannon mutual information identities, native `Transpose.TM` symbolic tree extractors, dummy-tape padding, and conditional closures of (T).

Unlike manuscripts with written-only sections, *every* numbered theorem, lemma, proposition, corollary, and example in this paper is completely proved in Lean 4, including the explicit 3-work-tape and 2-work-tape `Transpose.TM` transition-table constructions for Stone's shuffle, Stone's unshuffle, and row shear (Proposition 6.6), the native symbolic causal-cone tree evaluators (Lemma 8.1 and Theorems 8.2 and 8.3), the explicit primitive 12-th root of unity $\zeta_{12} = \exp(2\pi i/12)$ in $\mathbb{C}$

(Example 5.8), and the finite $d = 4, 7, 12, 13$ sharpness and characteristic-split witnesses. Open questions ((T), the $M = 1, \Delta = O(1)$ half-barrier, the macroscopic two-stage rank target, the $d = 5$ three-term threshold, and the periodic Walsh multi-scale target) are defined as explicit ': Prop' declarations so that all logical equivalences and conditional implications among them are themselves machine-checked theorems. Where Proposition 2.5(a) formalizes the separated storage-code amplification benchmark from [10], its theorem statement retains the explicit odd-girth seed hypothesis `HasOddGirthLinearSeed` (matching `TapeStorageRates.OddGirthLinearSeed` of [10, Definition 5.1]).

B Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `BoundedArityDecoders`. A partial status retains the gap stated at its theorem.

`AffineCubeComposition.lean` `CauchySchwarzAffineCubeExtractionSpec`, `Control_D13TwoCubeAPFreeSharpnessSpec`, `Control_D4SameIntervalCodedFactorizationSpec`, `Lemma_AffineCubeCommonMessageGrowthSpec`, `Theorem2A_TwoStageRamseyFreeCutoffSpec`.

`AffineCubeCompositionProofs.lean` `cauchySchwarzAffineCubeExtraction`, `control_D13TwoCubeAPFreeSharpness`, `control_D4SameIntervalCodedFactorization`, `lemma_AffineCubeCommonMessageGrowth`, `theorem2A_TwoStageRamseyFreeCutoff`.

`AffineCubeConsequences.lean` `control_D13_GeneralFieldAndProgressionBridge`, `control_D4_AllWidthsAndBinaryFields`, `denseSubset_AffineCubeCommonMessageBound`, `singleSubset_CauchySchwarzAffineCubeExtraction`, `theorem2A_GrowingClassesNonFactorization`.

`BoundedAritySetCapacity.lean` `BoundaryControl_DeficitVanishesAtDeltaEqDSpec`, `ExistsRecoverableSubsetBoundSpec`, `GrowingClassesPositiveDeficitSpec`, `OneClassArityTwoHalfBarrierQuestion`, `OneClassBoundedArityCodingGainAboveHalfQuestion`, `OneClassBoundedArityHalfBarrierQuestion`, `OneClassBoundedArityInverseAritryScalingQuestion`, `OneHopRecoverableInjectiveSpec`, `SelfReferentialAndCleanCountSpec`, `Theorem1_BoundedAritySetCapacitySpec`.

`BoundedAritySetCapacityProofs.lean` `boundaryControl_DeficitVanishesAtDeltaEqD`, `existsRecoverableSubsetBound`, `growingClassesPositiveDeficit`, `oneClassBoundedArityCodingGainAboveHalf_iff_not_halfBarrier`, `oneClassCodingGain_of_inverseAritryScaling`, `oneHopRecoverableInjective`, `selfReferentialAndCleanCount`, `theorem1_BoundedAritySetCapacity`.

`DiagnosisAndBarriers.lean` `Diagnosis_AbstractSeparatedHighRateCodeSpec`, `Diagnosis_BehrendAndRandomAPBoundSpec`, `Diagnosis_LinearSharedDecoderBoundsAndMaskFloorsSpec`, `Diagnosis_RamseyHalesJewettTwoStageSpec`, `Diagnosis_ShiftAndRotationCodingGainBarrierSpec`, `Diagnosis_StoneShuffleAndRowShearMachineComplexitySpec`, `Diagnosis_TCPAndAdaptiveConeReductionSpec`.

`DiagnosisAndBarriersProofs.lean` `diagnosis_AbstractSeparatedHighRateCode`, `diagnosis_BehrendAndRandomAPBound`, `diagnosis_LinearSharedDecoderBoundsAndMaskFloors`, `diagnosis_RamseyHalesJewettTwoStage`, `diagnosis_ShiftAndRotationCodingGainBarrier`,

diagnosis_StoneShuffleAndRowShearMachineComplexity, diagnosis_TCPAndAdaptiveConeReduction, stoneUnshuffleTM_twoTape_linear.

DiagnosisAndBridgesConsequences.lean diagnosis2_HighRateTendstoOne, diagnosis3_MultiBitBlockEdgeShiftBarrier, diagnosis4_ExplicitSquareCoverAndMaskCertificateBridge, diagnosis5_CombinatorialInvVdWTendstoAndRankBound, diagnosis6_ModularBehrendAndRandomZModProgression, diagnosis_Paper18AdaptiveOriginalEntryPullback, tmMatrix_gridLinear_bridges, tmWholeCellCutConeBound, wordStepTrace_TCPCellConeReduction.

DualityAndSwapConsequences.lean antiDiagTensorEquiv_intertwines_rot, bonus1_UnifiedThreeTermExceedsPairingBound, bonus2_SublinearProfileAndTendstoHalf, bonus2_WalshDichotomyStructuralProperties, boundedOSRPipelinePassLowerBound, isCyclicInterval_cyclicBoundaryCount_le_two, isPeriodicScaleDyadicMask_cyclicBoundaryCount, isRowOrColUniform_rankLe_one, maskFloor_GenericSubspaceRealization, rowDiagonal_SourceArityGaugeInvariance, shiftSetCovariance_tensor_finrank, skewOneTimePad_ShannonMutualInfoZero, uniformShannonMI_eq_zero_of_indep, walshBitSet_cyclicBoundaryCount, walshBitSet_joint_separates_points, walshDigitMaskRotSpan_finrank_le.

MachineSweepConsequences.lean
conditional_NoLowErrorTwoStageLinear_of_MacroscopicTarget, conditional_TapeCodingConjecture_of_HalfBarrierAndOneClassFibre, conditional_TapeCodingConjecture_of_PeriodicCellSweepHypothesis, evalConeTree_traceConeTree_eq, padDummyWorkTape_transposesWithin, tmPeriodicSweepExtraction_cutAbsorbed, tmPeriodicSweepExtraction_cutLabelRefined.

NativeSweepInterface.lean PeriodicSweepInducesBoundedAritySharedCodeSpec, PeriodicSweepNonlinearFibreDeficitSpec, TMTraceCutConeDeterminationSpec, TapeCodingConjectureNegationSpec.

NativeSweepInterfaceProofs.lean entryBits_eq_clog, periodicSweepInducesBoundedAritySharedCode, periodicSweepNonlinearFibreDeficit, tmTraceCutConeDetermination.

OperatorDuality.lean AntiDiagonalTensorSeparationSpec, Control_RowShearAndRowDiagonalAlternationSpec, Control_SkewOneTimePadInformationTrapSpec, OperatorSchmidtRankStoneShuffleSpec, OperatorSchmidtRankSubmultiplicativeSpec, OperatorSchmidtRankSwapSpec, PairwiseShiftCollisionCharacterizationSpec, StoneShufflePipelinePassLowerBoundSpec.

OperatorDualityProofs.lean antiDiagonalTensorSeparation, control_RowShearAndRowDiagonalAlternation, control_SkewOneTimePadInformationTrap, operatorSchmidtRankStoneShuffle, operatorSchmidtRankSubmultiplicative, operatorSchmidtRankSwap, pairwiseShiftCollisionCharacterization, stoneShufflePipelinePassLowerBound.

SetCapacityConsequences.lean asymptotic_SublinearClassesBrokenFractionRefutesRateOne, canonicalWordAlphabet_card_and_ge_two, corollary_OneClassArityOneThreeQuarters, corollary_OneClassArityTwoTwentyThreeTwentySevenths, growingClasses_strict_inv_exp_deficit, oneClassCanonicalArityTwo_of_allAlphabetHalfBarrier, pMin_two_sided_theta_inv_arity,

theorem1_BoundedAritySetCapacity_unconditional, theorem1_CanonicalWordSetCapacity, wall1_ExactLogarithmicAsymptoticAllDimensions, wall1_UnrestrictedArityRateLowerBoundAndTendstoOne.

SumsetRankConsequences.lean control_D12_BandRowSpaceChannelEquiv, control_D12_UnconditionalExists, d12AddChar_injective_and_primitive, exists_primitive_twelfth_root_complex, finsetNatDiam_shift_min, sumsetAnnihilatingPolynomial_bandRowSpace, theorem2BExplicitConst_pos, theorem2B_ExplicitConstant, theorem2B_OneSecondStage_OmegaLog, unshiftedToShiftedSumset_intrinsicDiam.

SumsetRankDeficit.lean Control_D12SingleClassAbstractBandSpec, IteratedKovariSosTuranSumsetSpec, OpenTarget_MacroscopicTwoStageRankDeficitAdmissibleFieldSpec, OpenTarget_MacroscopicTwoStageRankDeficitSpec, StackedBandPullbackInjectiveSpec, SumsetAnnihilatingPolynomialSpec, SumsetTwoHopRankBoundSpec, Theorem2B_TwoStageLogPowerRankDeficitSpec, UnshiftedToShiftedSumsetSpec, VandermondeArbitrarySubsetInjectiveSpec.

SumsetRankDeficitProofs.lean control_D12SingleClassAbstractBand, iteratedKovariSosTuranSumset, openTarget_admissible_of_unrestricted, stackedBandPullbackInjective, sumsetAnnihilatingPolynomial, sumsetTwoHopRankBound, theorem2B_TwoStageLogPowerRankDeficit, unshiftedToShiftedSumset, vandermondeArbitrarySubsetInjective.

SwapCodeAnswers.lean Bonus1_ThreeTermBlockSwapCodeDimSpec, Bonus1_ThreeTermPaddedSevenAndFiveSpec, Bonus2_MultiscaleFullProfileAttainsFullSwapSpec, Bonus2_MultiscaleIntervalBothSidesSpec, OpenQuestion_ThreeTermDimFiveExceedsPairingSpec, OpenTarget_MultiscalePeriodicWalshProfileSpec, OpenTarget_MultiscalePeriodicWalshSwapCodeSpec.

SwapCodeAnswersProofs.lean bonus1_ThreeTermBlockSwapCodeDim, bonus1_ThreeTermPaddedSevenAndFive, bonus2_MultiscaleFullProfileAttainsFullSwap, bonus2_MultiscaleIntervalBothSides, openTarget_walsh_of_periodicProfile, walshDigitMask_in_periodicDyadicSpan.

Other declarations named in the text CanonicalWordAlphabet, CausalConeTree, GridState, HasBarrelShiftCircuit, HasCausalCompleteExchangeCode, HasExactOperatorSchmidtRank, HasOddGirthLinearSeed, HasOperatorSchmidtRankLe, HasOutputRowSourceArityLe, InAlignedDyadicIntervalRotSpan, InCutCausalCone, InMultiscaleIntervalRotSpan, InMultiscalePeriodicDyadicRotSpan, InPairwiseShiftCovariance, IsAlignedDyadicInterval, IsBoundedAritySetCode, IsColUniformRowOp, IsCyclicInterval, IsFrozenTraceCutFibre, IsPeriodicScaleDyadicMask, IsRowDiagonalOp, IsRowUniformColOp, IsThreeByThreeSymmetricZeroTraceBlocks, OneClassSweepFibreReductionHypothesis, PeriodicCellSweepReductionHypothesis, SlotConeTree, SyntacticCellSupport, SyntacticColumnSupport, ThreeAPFreeMod, WordStepFrozenTrace, alternatingRowDiagProduct, antiDiagLinear, antiDiagTensorEquiv, arityGainFactor, bernoulliSubsetWeight, ceilSqrtTwoMul, charBandEval, cleanEntries, colBitFlipGridLinear, composeGridOps, cutFreeTraceConeTree, cutTouchingOutputCells, cyclicBoundaryCount, cyclicChannelShift, encodeTMMatrixToGrid, evalConeTree, finsetNatDiam, fromAntiDiagLinear, gridSwapLinear, halfIntervalPartition4, leftComponentSubmodule, orderedCubeMap, outputBitIdx, recoverableCells, rowShearLinear, rowShearTMMatrix, shiftSetCovarianceSubmodule, shiftedFootprint, skewXorPad,

stoneShuffleGridLinear, stoneShuffleTMMatrix, stoneUnshuffleGridLinear, stoneUnshuffleTMMatrix, theorem2BExplicitConst, theorem2BExplicitThreshold, threeTermBlockA, threeTermBlockB, threeTermPaddedA, threeTermPaddedB, traceConeTree, traceSlotConeTree, uniformShannonMI, walshBitSet, walshDigitMaskRotMatrix, wordStepTCPDepth.

C Building and verifying

The repository must be cloned together with its sibling paper directories so that relative Lake path dependencies resolve. The Lean package pins Lean 4 and Mathlib to version `4.34.1`. From the paper directory `Bounded-Arity-Shared-Decoders-and-Ramsey-Free-Composition-for-Matrix-Transposition`, run:

```
python3 tools/lean_index.py
cd lean
lake exe cache get
lake build
lake env lean PaperAxioms.lean
cd ../paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Running `python3 tools/lean_index.py` regenerates `lean/BoundedArityDecoders.lean`, `lean/PaperAxioms.lean`, and Appendix B (`paper/sections/leanindex.tex`) directly from the `\formal` and `\lean` commands in `paper/sections/*.tex`. Every audited declaration in `PaperAxioms.lean` depends at most on the three standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`; there are no `sorry`, `admit`, `native_decide`, `unsafe`, `implemented_by`, `extern`, or custom `axiom` declarations anywhere in the package. Some finite algebraic and combinatorics proofs set local `maxHeartbeats` allowances for elaboration, which do not affect statement types or kernel verification.

References

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